

## Part II

# Characterization

Chapter 8

Surface Analysis Methods for Contaminant Identification

David A. Cole<sup>1</sup>, Sachin Attavar<sup>1</sup> and Lei Zhang<sup>2</sup>

<sup>1</sup>Evans Analytical Group, East Windsor, NJ, USA, <sup>2</sup>DuPont Central Research and Development, Wilmington, DE, USA

Chapter Outline

|                                                                   |            |                                                                        |            |
|-------------------------------------------------------------------|------------|------------------------------------------------------------------------|------------|
| <b>8.1 Introduction</b>                                           | <b>333</b> | 8.4.1 Background of TOF-SIMS                                           | 365        |
| <b>8.2 Auger Electron Spectroscopy</b>                            | <b>334</b> | 8.4.2 Basic Principles of TOF-SIMS                                     | 366        |
| 8.2.1 Background of AES                                           | 334        | 8.4.3 TOF-SIMS Instrumentation                                         | 371        |
| 8.2.2 Basic Principles of AES                                     | 336        | 8.4.4 Applications of TOF-SIMS for Characterizing Surface Contaminants | 372        |
| 8.2.3 AES Instrumentation                                         | 341        | 8.4.5 Recent Developments and Future Directions of TOF-SIMS            | 376        |
| 8.2.4 Applications of AES for Characterizing Surface Contaminants | 341        | <b>8.5 Low-Energy Ion Scattering</b>                                   | <b>378</b> |
| 8.2.5 Recent Developments and Future Directions of AES            | 346        | 8.5.1 Background of LEIS                                               | 378        |
| <b>8.3 X-Ray Photoelectron Spectroscopy</b>                       | <b>347</b> | 8.5.2 Basic Principles of LEIS                                         | 378        |
| 8.3.1 Background of XPS                                           | 347        | 8.5.3 LEIS Instrumentation                                             | 383        |
| 8.3.2 Basic Principles of XPS                                     | 348        | 8.5.4 Applications of LEIS for Characterizing Surface Contaminants     | 384        |
| 8.3.3 XPS Instrumentation                                         | 356        | 8.5.5 Recent Developments and Future Directions of LEIS                | 387        |
| 8.3.4 Applications of XPS for Characterizing Surface Contaminants | 357        | <b>8.6 Summary</b>                                                     | <b>387</b> |
| 8.3.5 Recent Developments and Future Directions of XPS            | 364        | <b>Note</b>                                                            | <b>388</b> |
| <b>8.4 Time-of-Flight Secondary Ion Mass Spectrometry</b>         | <b>365</b> | <b>References</b>                                                      | <b>388</b> |

8.1 INTRODUCTION

Contamination in the broadest sense affects nearly all manufacturing industries. As such, it is virtually impossible to present a detailed review of all analytical

methods used for contaminant identification. However, if one takes a subset of contaminants, namely only those at surfaces, the task is considerably simplified. As industries push for detection of sub-monolayer films and sub-micrometer particles, the number of generally applicable surface analysis methods becomes quite limited. Four methods were selected for inclusion in this chapter: Auger electron spectroscopy (AES), X-ray photoelectron spectroscopy (XPS), time-of-flight secondary ion mass spectrometry (TOF-SIMS), and low-energy ion scattering (LEIS). These methods in combination are fully capable of examining sub-monolayer films and sub-micrometer particles, and are readily available through numerous industrial, academic, and national laboratories.

As will be demonstrated, each method is best suited for specific contaminant studies. Judicious selection of a particular method requires knowledge of substrate and contaminant properties, such as size, shape, morphology, composition, volatility, concentration, and history. The selection process often begins with inspection by optical or electron microscopy to estimate whether the contamination is due to a surface film or particles. In addition to this, one must decide whether the study should provide qualitative or quantitative results and what specific type of information is required, for example, elemental composition, chemical bonding, or exact compound identification. The answers to these questions will generally limit the choice to one or two methods. Additional criteria for method selection include: detection limit, sampling depth, imaging, and depth profiling. These properties are summarized for the four methods in [Table 8.1](#). For a detailed description of the basis for these properties, see the individual sections on basic principles. For a detailed description of how these techniques have been employed to study contaminants on contemporary products, see the individual sections on applications.

## 8.2 AUGER ELECTRON SPECTROSCOPY

### 8.2.1 Background of AES

AES is one of a handful of techniques routinely used for surface analysis. It has reached this position because it can quantitatively detect all elements except hydrogen and helium, in the outer 10 nm of a sample. The technique is based on the fact that when a core electron is removed from an atom, the core electron vacancy can be filled by an electron from an outer (lower energy) shell with the consequent emission from the atom of a second outer shell electron. The “Auger process” and the “Auger” electron are named after Pierre Auger, who reported the phenomenon in 1925.<sup>1</sup> However, not until 1953 was it shown that Auger electrons, produced by bombarding a solid with electrons, could be used for surface analysis.<sup>2</sup> Then in 1968 the first practical Auger electron detection scheme was reported.<sup>3,4</sup> This set the stage for explosive development of the technique in the 1970s and 1980s and more modest growth in the past 15 years.

**TABLE 8.1** Overview of Selected Properties for Four Surface Analytical Methods

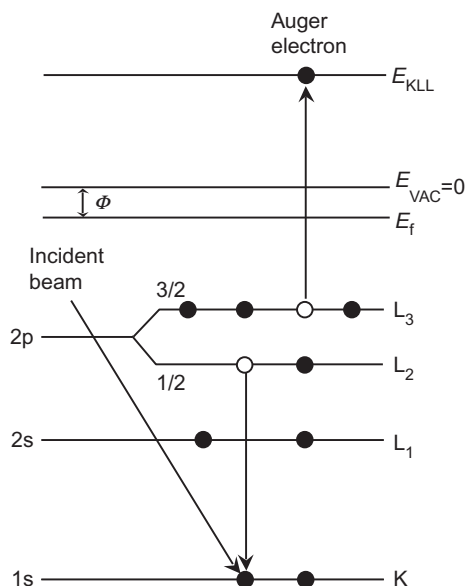
| Method                | AES                                                                | XPS                                                              | TOF-SIMS                                                                                  | LEIS                                                                         |
|-----------------------|--------------------------------------------------------------------|------------------------------------------------------------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|
| Material              | Conducting or semiconducting solids. Insulators are very difficult | Any solid                                                        | Any solid                                                                                 | Any solid                                                                    |
| Information           | Elemental. Oxidation state or chemical bonding in select cases     | Chemical bonding, oxidation state, and elemental                 | Molecular weight, chemical bonding, and elemental                                         | Elemental                                                                    |
| Quantitative analysis | Yes, best with standards                                           | Yes, best with standards                                         | Only with standards                                                                       | Yes, standards not required                                                  |
| Minimum analysis size | 10 nm diameter                                                     | 10 $\mu\text{m}$ diameter                                        | $\sim 2 \mu\text{m}$ diameter for organic analysis, 50 nm diameter for inorganic analysis | 100 $\mu\text{m}$ diameter                                                   |
| Sampling depth        | 5–25 monolayers                                                    | 5–25 monolayers                                                  | 1–3 monolayers                                                                            | 1 monolayer                                                                  |
| Detection limit       | 0.1–1.0 at. %                                                      | 0.01–1.0 at. %                                                   | 1 ppma, $1 \times 10^8$ atom/ $\text{cm}^2$                                               | $1 \times 10^9$ to $5 \times 10^{15}$ atom/ $\text{cm}^2$                    |
| Imaging               | Yes, rapid                                                         | Yes                                                              | Yes, rapid                                                                                | Yes, in principle                                                            |
| Depth profiling       | Yes, rapid                                                         | Yes                                                              | Yes, rapid                                                                                | Yes                                                                          |
| Major limitations     | Electron beam damage. Charging of insulating samples               | Relatively large analysis area. Analysis is often time consuming | Ion yields vary by orders of magnitude. Standards needed for quantitative analysis        | Poor sensitivity for low-mass atoms. Poor resolution between high-mass atoms |

### 8.2.2 Basic Principles of AES

To understand how AES can be used to study contaminants on surfaces, one needs to examine the Auger process more closely. For demonstration purposes consider a piece of household aluminum foil. The bulk of the foil is metallic aluminum; however, the outermost surface consists of organic contaminants such as rolling oils, and a native aluminum oxide layer that always forms when metallic aluminum is exposed to air. When a high-energy electron beam bombards this sample, core electrons are removed from atoms of all types within a depth of around 1  $\mu\text{m}$ . For aluminum atoms the core electron is the K or 1s shell, having energy  $E_x$  (see Figure 8.1). Ionization leaves the atom in an excited state from which it can partially relax by filling the core vacancy with an electron from an outer shell, in this case the  $L_{2,3}$  or 2p, having energy  $E_y$ . The energy difference between the K and  $L_{2,3}$  shells,  $E_x - E_y$ , is transferred to a second electron, say from the  $L_{2,3}$  shell, with initial energy  $E_z$ , thereby causing it to be ejected from the atom. The kinetic energy of the emitted electron,  $E_A$ , is the transferred energy,  $E_x - E_y$ , minus the initial energy of the emitted electron,  $E_z$ , that is

$$E_A = E_x - E_y - E_z \quad (8.1)$$

Because the energy of each shell is specific to the atom, the kinetic energy of the emitted electron is also specific to the atom from which it originated. Hence an Auger spectrum will have at least one peak for every atom having at least



**FIGURE 8.1** Schematic diagram of the Auger electron emission process. (Source: Evans Analytical Group, Sunnyvale, CA. Reprinted with permission.)

three electrons. Hydrogen and helium, having less than three electrons, cannot undergo the Auger process.

Auger electrons are typically referred to by the letters of the shells involved in the Auger process, that is, Al KL<sub>2,3</sub>L<sub>2,3</sub> in the example above. After this Auger process the atom is left in a doubly ionized state with two missing L-shell electrons. These vacancies can be filled by electrons from the valence band with transference of energy to other valence electrons, which are ejected from the atom. This creates a second peak in the spectrum due to Al L<sub>2,3</sub>VV electrons. Clearly, as the number of electrons in an atom increases, so will the number of peaks in an Auger spectrum. Fundamentally an Auger spectrum thus consists of a plot of the number of electrons emitted versus electron kinetic energy. In this plot there will be one or more peaks for every type of atom in the sampling volume. Typically, AES refers only to spectra generated by bombardment of the solid with high-energy electrons. However, it is important to note that the Auger process will take place whenever a core vacancy is generated. Hence “Auger” electrons are generated when a sample is irradiated with X-rays as in XPS, also known as electron spectroscopy for chemical analysis (ESCA), or in some cases when irradiated with ions such as when carrying out sputter depth profiling.

Since AES relies on measuring the kinetic energy of electrons emitted from the atom, the electrons must actually be emitted not only from the atom but also from the solid itself. This has a profound effect on the sampling depth in AES. In a typical AES measurement the primary electron beam would have an energy of 3–20 keV. The high energy of the primary electrons causes Auger electrons to be generated at depths up to  $\sim 1\ \mu\text{m}$ . As these electrons travel through the sample they undergo inelastic collisions which reduce their kinetic energies. Since the degree of energy loss is not quantized, inelastically scattered electrons appear as part of the loss tail at the low-energy side of the Auger peak. Only those electrons that exit the solid without experiencing inelastic collisions contribute to the main body of the Auger peak. Inelastic collisions are random chance events and as such the inelastic mean free path,  $\lambda$ , is defined as the distance electrons travel after which 27% will have lost energy and 63% will still have their initial energy. To a first approximation, the inelastic mean free path has been found to be roughly proportional to the square root of the kinetic energy of the electrons and typically ranges between 0.5 and 2.5 nm for kinetic energies of 100 and 2000 eV, respectively. For the Al L<sub>2,3</sub>VV electrons in our example, which have an energy of  $\sim 50\ \text{eV}$ , 63% will have come from the  $1\lambda$  ( $\sim 0.5\ \text{nm}$ ), 95% come from  $3\lambda$  ( $\sim 1.5\ \text{nm}$ ), and 99% come from  $5\lambda$  ( $\sim 2.5\ \text{nm}$ ). In contrast, for Al KL<sub>2,3</sub>L<sub>2,3</sub> electrons having energies of  $\sim 1380\ \text{eV}$ , 63% come from  $\sim 2\ \text{nm}$ , 95% come from  $\sim 6\ \text{nm}$ , and 99% come from  $10\ \text{nm}$ . Clearly, AES is extremely surface sensitive. Unfortunately, inelastic scattering introduces one of the more confusing issues in AES, namely sampling depth. In fact, there is no single AES sampling depth; it depends on the atom of interest, the specific electron monitored, and how one defines “sampling,” that is, 63%, 95%, and 99% of the signal, or some other “definition.”

With an understanding of sampling depth, it is an almost trivial task to understand AES sampling volumes. AES uses high-energy primary electron sources similar to those found in scanning electron microscopes. These electron beams are very easy to steer to discrete locations on a surface or to raster across a surface. Thus, changing the size and shape of the sampling volume is a simple task. For instance, if the primary beam is focused to a fine point, say 10 nm in diameter, the sampling volume would take the shape of an exceedingly small sphere roughly 10 nm in diameter. Alternatively, if the primary electron beam were rastered over an area, say  $100 \times 100 \mu\text{m}^2$ , the sampling volume would take the shape of a very thin square box. This wide range of sampling volumes is ideally suited for analysis of contaminants such as films and sub-micrometer particles.

The use of high-energy electrons to create core vacancies dictates that Auger spectra can be obtained from conductors and semiconductors, and also from insulating films less than a few micrometers thick on conducting substrates. Bulk insulators are exceedingly difficult, if not impossible, to analyze by AES. The technique of coating insulating samples with conducting layers, as is often done prior to energy dispersive spectroscopy (EDS) analysis, generally does not work in AES due to the shallow sampling depth. A side benefit of using high-energy electrons is that most AES instruments are capable of secondary electron imaging. Some commercial systems are capable of additional imaging methods such as backscattered electron and absorbed current imaging. Thus, the location and morphology of surface features, including contaminants, can be documented prior to elemental analysis by AES.

From the description so far, one can see that AES can identify the presence of all elements except H and He in the first few atomic layers. That in itself would be useful, but AES is also a quantitative technique. The traditional method of quantifying elements is through the use of peak-to-peak heights in the first-derivative spectrum. The reasons for using first-derivative spectra will be discussed later in this section. One peak each is chosen for each of the elements present. Barring spectral interferences, this is generally the most intense peak for that element. The peak-to-peak heights are then normalized by dividing by the elemental sensitivity factor. The normalized values are then expressed as a percentage of the total with the units being atom percent. Commercial instruments come with tables of sensitivity factors that can be used for unknown materials, such as in contamination studies, with an accuracy of 50-80%. Quantitative accuracy of >95% requires calculation of sensitivity factors specific to a given material. Moreover, this material must be of constant composition within the sampling volume.

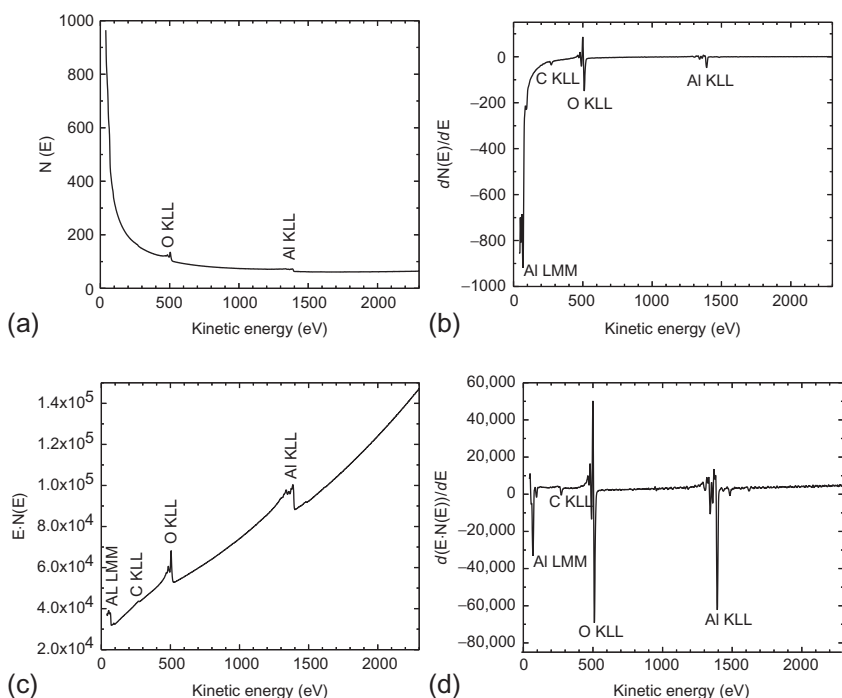
There are six dominant factors affecting sensitivity factors. These are (1) the probability of forming the initial vacancy, (2) the probability that the initial vacancy will be filled by an outer shell electron, (3) the probability that the filling of the core vacancy will cause emission of an Auger electron, (4) the magnitude of inelastic scattering in the solid, (5) the detection efficiency of the

spectrometer, and (6) peak shape. Factor 1, the probability of forming the initial vacancy, is controlled by the energy of the primary electron beam. It has been shown that the maximum production of vacancies requires a primary energy 3-6 times the energy of the core shell. Thus, for the Al LVV peak, where the energy of the L shell is less than 0.1 keV, the highest sensitivity beam energy would be <1 keV. Conversely, for the Al KLL peak, where the energy of the K shell is around 2 keV, the highest sensitivity beam energy would be 5-10 keV. Factors 2, 4, and 6 are controlled by the composition of the solid but vary only slightly with composition. To a first approximation, they are assumed to be constant. Factor 5, the detection efficiency of the spectrometer, is fixed by the instrument design. And finally, factor 3, the probability that the filling of the core vacancy will cause emission of an Auger electron, is a fundamental property of the atom and is therefore constant.

Although the probability that filling of the core vacancy will cause emission of an Auger electron is constant, it is worth investigating in greater detail. There are actually two competing processes for returning the atom to its initial state: emission of an Auger electron and emission of a characteristic X-ray, that is, X-ray fluorescence. Since two pathways exist, the probability of Auger electron emission is unity minus the probability of X-ray emission. Auger emission is the highly favored process for filling core vacancies having energies less than about 2 keV. Thus, Auger emission is the dominant process for K-shell vacancies of boron through arsenic. K-shell vacancies in elements heavier than arsenic relax primarily via X-ray emission. Although this sounds like AES would be insensitive to heavy elements, in fact other highly probable Auger emissions exist for vacancies from other shells, that is, L, M, and N. Fortunately, one can select peaks for all elements in which the Auger emission to X-ray emission ratio is >0.9. Because of this, AES typically is well suited for analysis of unknown materials, particularly those suspected of containing light elements such as boron through sodium.

The discussion above has alluded to the fact that Auger spectra can be plotted as the number of emitted electrons,  $N(E)$ , versus electron kinetic energy,  $E$ , or as a first derivative  $dN(E)/dE$  versus  $E$ . Actually there are many ways to display the data. The most common methods are not based on properties intrinsic to Auger electrons but on the large number of background electrons. At low energies, <300 eV, the background level of electrons emitted from a solid is dominated by low-energy secondary electrons, often called “true” secondary electrons. At higher energies, the background is dominated by high-energy secondary electrons and, to a much lesser degree, by Auger loss tails. Superimposed on this background are the Auger electrons of interest, yielding a plot of  $N(E)$  versus kinetic energy,  $E$  [see Figure 8.2(a)]. By taking the first derivative of this, one can remove much of the influence of the high-energy secondary electron background [see Figure 8.2(b)]. However, the low-energy secondary electrons still disproportionately affect the low-energy region. To overcome this, one can multiply the number of electrons by any function that



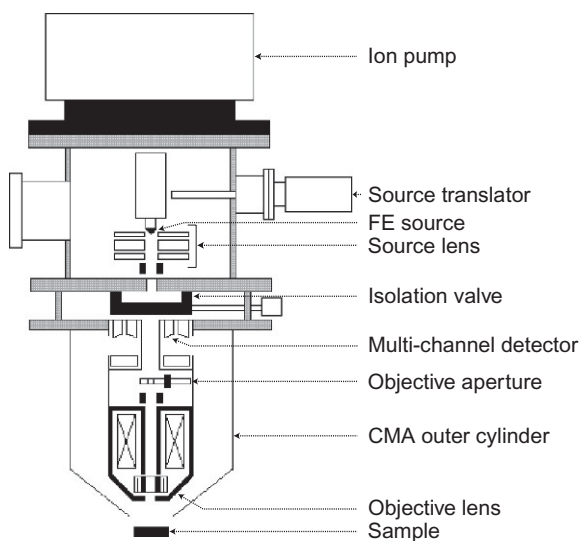


**FIGURE 8.2** AES spectra of aluminum foil with signal intensity displayed as: (a) the number of emitted electrons, (b) the first derivative of the number of emitted electrons, (c) the electron's kinetic energy times the number of emitted electrons, and (d) the first derivative of the electron's kinetic energy times the number of emitted electrons. (Source: Evans Analytical Group, Sunnyvale, CA. Reprinted with permission.)

increases with electron kinetic energy, for example, the electron kinetic energy itself. This yields the  $E \cdot N(E)$  versus  $E$  plot shown in Figure 8.2(c). The first derivative of this,  $d(E \cdot N(E))/dE$  versus  $E$ , removes most of the secondary electron influence on the spectrum leaving sharp Auger electron peaks [Figure 8.2(d)]. The last figure is in fact the historic way Auger spectra have been displayed since  $dE \cdot N(E)/dE$  versus  $E$  is the nominal output of many older detection schemes. In this form quantitative analysis is performed, as stated earlier, from the peak-to-peak height for a given electron emission. The first-derivative form,  $dE \cdot N(E)/dE$  versus  $E$ , has the advantage of easy comparison with historic spectra, easy identification of low-intensity Auger peaks, and little user intervention needed for quantitative analysis. In contrast, the underivatized form,  $E \cdot N(E)$  versus  $E$ , has the advantage of better signal-to-noise, better definition of peak shape, and the potential for more accurate quantitative analysis. For most studies of unknown materials and thin-film contaminants, the first-derivative form is sufficient. However, for sub-micrometer particle analysis, the underivatized form is gaining popularity due to the better signal-to-noise. This is particularly true when qualitative particle screening is required rather than quantitative analysis.

### 8.2.3 AES Instrumentation

All Auger spectrometers consist of a vacuum chamber operated at a pressure less than  $10^{-8}$  torr housing a high-energy electron source and an electron energy analyzer, often co-axial to each other (see Figure 8.3). Other common features are sample introduction and manipulation hardware, a secondary electron detector for obtaining secondary electron images, and inert gas ion guns for sample cleaning and depth profiling. Cooling and heating stages are useful for preservation of volatile components and sample modification, respectively. Current commercial instruments permit computer control of most components. Computer control of the high-energy electron source simplifies analysis at a given point on the sample such as an individual particle, or a specific area such as a film. Combined control of the source and detector permits acquisition of elemental (or in special cases, chemical) images and line-scans, that is, a display of peak intensity versus location on the sample. Of particular importance is the ability to digitally store spectral data for subsequent analysis such as quantification, curve fitting, factor analysis, and principal component analysis.



**FIGURE 8.3** Schematic diagram of an AES chamber. (Source: Physical Electronics Inc., Chanhassen, MN. Reprinted with permission.)

### 8.2.4 Applications of AES for Characterizing Surface Contaminants

An ideal contaminant identification tool would provide elemental composition, chemical bonding, and exact compound identification of organic, inorganic, and metallic compounds of any size and any phase—solid, liquid, or gas. Unfortunately, such a tool does not exist. However, AES fulfills many of these

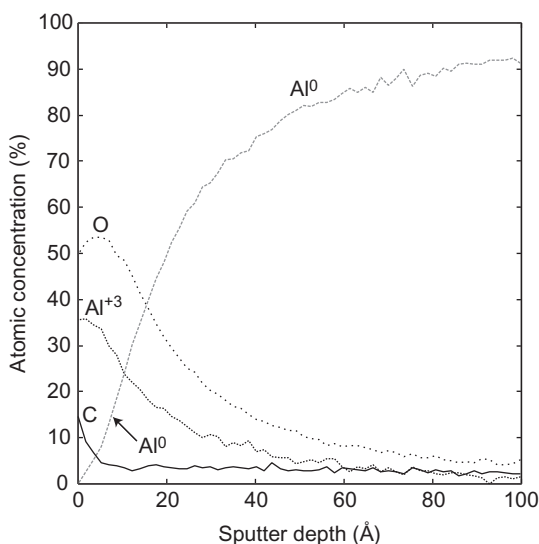
requirements. If we momentarily limit contaminants to those existing on surfaces, then the task at hand would be to analyze particles and films, though not necessarily continuous films. Since particles and films can be very small in one or more dimensions, a surface-sensitive method would be useful. Auger electrons, by the fact that they have low energies, must come from a depth no greater than  $\sim 10$  nm. Moreover, all elements except for hydrogen and helium yield Auger electrons and the number of electrons emitted for each element is directly proportional to the atomic concentration of that element. Hence, AES is a quantitative surface analysis method. In most cases Auger peak energies are sufficient for identifying the element from which the electron originates, but not the chemical or oxidation state of that atom. Therefore, compound identification by AES is deduced from the relative abundance of elements detected. For example, if a particle was found to contain only boron and nitrogen in roughly equal concentrations, it would be safe to describe the particle as boron nitride. Of course, when many elements are detected at the same location, compound identification is more difficult.

Because AES uses a high-energy electron beam as the probe, it is limited to analysis of conducting and semiconducting surfaces, or to insulating films and particles on conducting or semiconducting substrates. Note that a common restriction for insulating films and particles is that at least one dimension should be less than a few micrometers. The use of an electron beam probe has the additional benefit that the sample can be viewed by a secondary electron image. This permits the operator to inspect and document surface morphology, and to perform analysis on discrete features such as individual particles. When secondary electron contrast is insufficient to locate the desired feature, one can perform survey analysis from an area encompassing the suspected contaminant and then map the detected elements. Auger maps are two-dimensional pseudo-color images showing the intensity of a given emitted electron as a function of  $X$ ,  $Y$  location on the surface. If the elemental composition of the contaminant differs significantly from the substrate, one or more maps should clearly identify its location. Selected area analysis can then be used to evaluate the contaminant in detail. For complex assemblies of contaminants this sequence of steps may need to be iterated several times.

If the contaminant of interest is not on the outer surface, then the overlying material must be removed. In the simplest case this is done by fracturing the sample, thereby exposing contaminants at the fracture face. This method is quite useful when studying adhesion failure, weld failure, and fatigue cracking. If sample fracturing does not reveal the contaminant, then the overlying material must be removed by chemical or mechanical means. Chemical etching is most useful when the etch process stops near but not actually at the contaminant since the process tends to deposit its own contaminants on the sample. The overlying material remaining after chemical etching is then removed by ion beam milling. Mechanical removal can be done in two ways: cutting and polishing, and ion beam milling. Cutting and polishing is acceptable for particles  $>1\ \mu\text{m}$  in

diameter and contaminant films  $>1\ \mu\text{m}$  thick. Since this is a rather small subset of typical contaminants, it is not widely performed. Conversely, ion beam milling is exceedingly common. The most common ion beam milling method uses an inert gas ion gun to remove material. By cycling between AES data acquisition and ion milling, a plot of concentration versus sputter time can be generated. Figure 8.4 shows a typical depth profile of household aluminum foil that has two contaminant layers, an outer organic layer  $\sim 0.3\ \text{nm}$  thick and a native oxide roughly  $2.5\ \text{nm}$  thick. If the sputter rate of the material is known, the plot is displayed as concentration versus sputter depth. A depth profile plot therefore shows the location, thickness, and composition of a contaminant layer. For buried particles, ion milling is often used solely to remove the overlayer at which time the exposed particle is analyzed as if it were a surface feature. Some of the newer Auger systems incorporate a focused ion beam (FIB) source for ion milling. These sources operate at very high current and very small beam diameters, allowing one to selectively ion mill trenches that bisect buried particles.

As one might expect, AES is widely used by the metallurgical community.<sup>5</sup> The ability to detect unknown elements at sub-monolayer coverage with high spatial resolution is vital to the study of interfacial failure. This includes adhesion failure in composites and adhesive joints, and fatigue failure of metals and alloys. In each case, segregation of contaminants at the interface produces a weak layer in which failure propagates. In contamination-induced adhesion failure in composites and adhesive joints, contamination often arises from improper/inadequate cleaning of the metal substrate prior to bonding. In fatigue

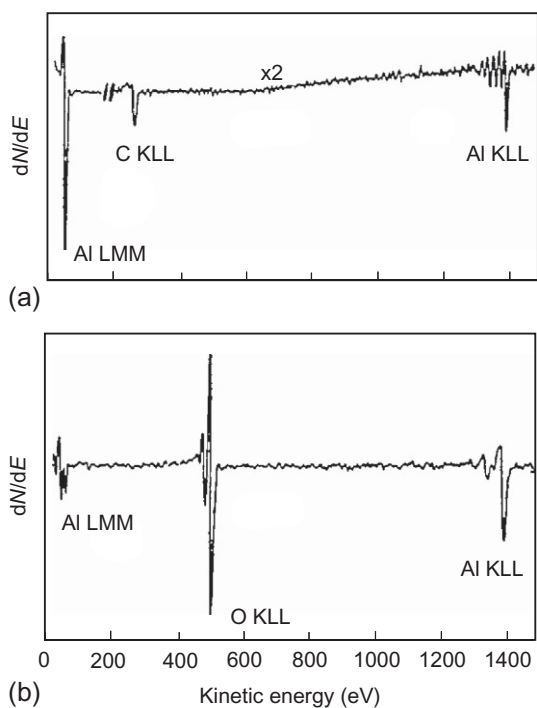


**FIGURE 8.4** AES depth profile of aluminum foil with a 0.3-nm-thick organic layer and a 2.5-nm-thick native oxide layer. (Source: Evans Analytical Group, Sunnyvale, CA. Reprinted with permission.)



failure of metals and alloys, the contaminants are generally present when the material is cast. An exception to this is weld failure, where contaminants both within and on the surface of the material can cause failure. Moreover, the contaminant can be introduced during welding. For example, Al-Li Alloy 2195 is a lightweight, high-fracture-toughness alloy that replaced much of the Al-Cu Alloy 2219 in the Super Light Weight Tank (SLWT) of NASA's Space Shuttle. AES investigations of test welds indicated that weld integrity was directly related to the amount of aluminum oxide at the grain boundaries at the weld site.<sup>6</sup> The presence of lithium increased the oxidation kinetics of the alloy such that low concentrations of oxygen in the shielding gas cause thin oxide film growth not seen in lithium-free alloys (see Figure 8.5).

Surface contamination of finished metal products is also an area of great interest. The most common detrimental result of such contamination is corrosion. Although one might have expected AES to have been used long ago in corrosion studies, it has in fact been routinely employed only in the past 15 years.<sup>5</sup> The complexity of corrosion processes and the desire to monitor chemical states rather than elemental identification has slowed the adoption of AES in this field. Nonetheless, AES is often used to study surface contaminants on metals. For example,

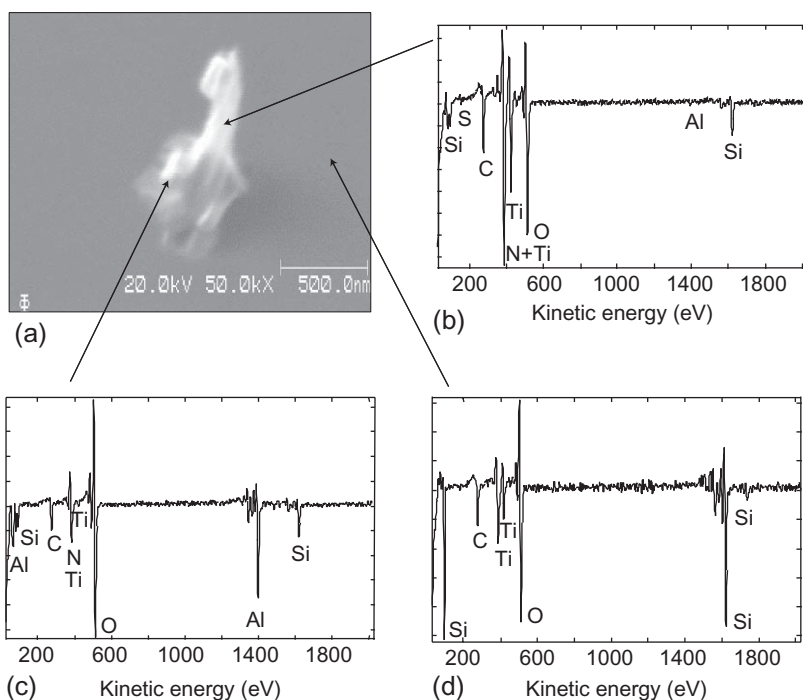


**FIGURE 8.5** AES spectra of fracture surfaces from: (a) a typical weld in Al-Cu Alloy 2219 and (b) a repair weld in Al-Li Alloy 2195. Note that both samples were vacuum fractured at  $1 \times 10^{-9}$  torr. (Adapted from Ref. 6. Reprinted with permission.)

blemishes in the cosmetic appearance of metal surfaces are often caused by surface contaminants. In other instances, surface contamination can directly affect the end-use properties. This is certainly the case for orthopedic implants. Many studies have been done to determine appropriate surface chemistry for biocompatibility. In such studies one should remember to include real-world issues such as sterilization. For example, a study of titanium implants sterilized by three different methods found considerable differences in particulate contamination and oxide growth.<sup>7</sup> Higher levels of contamination and oxide growth correlated with decreased cell attachment and spreading in assay studies, which may pose serious biological concerns. Thus, although the alloy composition and manufacturing process could produce an implant with excellent biocompatibility, it would be to no avail if improperly sterilized.

The ability to characterize sub-micrometer features and particles makes AES quite useful to the disk-drive industry.<sup>8</sup> The semiconductor and microelectronics industries also rely on AES for contaminant analysis. Most studies fall into two categories: process development and failure analysis. Many processes can and have been improved with the information provided by AES. Plasma etching is an excellent example. Plasma etching is an inherently dirty process because etch products either deposit on or attack chamber walls and hardware. By studying the composition of particles deposited on wafers during plasma etching, such as is shown in Figure 8.6,<sup>9</sup> one can determine their production mechanisms and thereby adjust process conditions and hardware as needed to minimize their production.<sup>9,10</sup> Whereas process development can be done in a controlled fashion, failure analysis is, by definition, uncontrolled. Contamination is often suspected when failure is related to adhesion, such as delamination of a multilayer metal stack or wire debonding. Delaminated metal stacks can be examined by depth profiling down to the failure layer, or more optimally by analyzing the two mating faces after delamination. Although the latter method results in slight surface contamination during sample preparation, it is often the preferred method. Wire debonding resulting from organic contamination or corrosion of the bond pad is easily studied by a combination of surface analysis for contaminant identification and depth profiling for contaminant thickness.<sup>11</sup>

AES is often used to evaluate the efficacy of a cleaning process and to measure consequential substrate damage. All high surface free energy materials are contaminated with airborne organic contaminants and most often with native oxide films. Since these contaminants often interfere with device operation, they must be removed during production. This is particularly true in the semiconductor industry. Different materials (Ge, AlN, GaN, InAs, etc.) may require different cleaning methods. However, in each of these cases AES can be used to monitor cleaning efficacy.<sup>12–14</sup> Care must be taken to assure that the substrate is not damaged during the cleaning process. AES analysis of several laser-cleaned substrates showed that organic contaminants could be removed without surface damage,<sup>15,16</sup> but removal of the native oxide from silicon caused permanent



**FIGURE 8.6** SEM micrographs of (a) a Ti/Al composite particle found on a silicon wafer after W etch back (WEB) processing, and AES spectra of (b) a titanium-rich region, (c) an aluminum-rich region, and (d) the wafer substrate. (Adapted from Ref. 9. Reprinted with permission.)

surface damage.<sup>17</sup> Other cleaning methods such as  $\text{Ar}^+$  bombardment<sup>18</sup> and  $\text{Ar}/\text{H}_2$  plasma treatment<sup>19</sup> can clean surfaces but may damage the cleaned surface or cause collateral damage to adjacent components.

### 8.2.5 Recent Developments and Future Directions of AES

As with many mature analytical techniques, recent advances in AES have been evolutionary rather than revolutionary. The three most important advances have been in speed and ease of analysis, analytical beam size, and sample handling. Many of the improvements in speed have been made possible by advances in electronics, multi-channel detectors, computers, and software. Faster electronics have the obvious effect of reduced dead time, and multi-channel detectors permit parallel electron detection. Faster and more powerful computers have made it much easier to operate Auger spectrometers. However, the largest impact from improved software and computing power has been in data analysis. Software features such as curve fitting, factor analysis, and principal component analysis, particularly when combined with underivatized spectra,  $E \cdot N(E)$  versus  $E$ , permit extraction of much more chemical information from AES spectra.

For the highest degree of chemical information, the use of an electrostatic hemispherical analyzer is recommended.

AES has always been an excellent tool for particle analysis due to the ability to confine the analysis to the particle and not the substrate upon which it resides. Particle analysis has been greatly improved by the incorporation of Schottky field-emission electron sources. These sources can be operated at extremely narrow beam diameters and at high beam currents, permitting AES analysis of features as small as 10 nm. However, this ability comes with a price, specifically the difficulty in preventing contamination of the sample with extraneous particles and in locating nanometer-sized particles. A solution to these problems has been found in the semiconductor industry.<sup>20</sup> Extraneous particles come from two sources: sample preparation such as cleaving the wafer to a suitable analytical size, and sample handling. Design of instruments capable of handling whole 200- and 300-mm wafers under cleanroom conditions eliminates particle generation during sample preparation and handling. Stand-alone optical defect detection (ODD) tools based on laser light scattering can be used to map and count particles on whole wafers. However, to date, ODD tools have not been incorporated into Auger spectrometers. Rather than requiring the operator to hunt for particles, a daunting task if one is looking for 10-nm particles, automated stage navigation is available using imported files from ODD tools. While the operator must still make the final selection of the analysis location, the particle of interest is typically in the field of view of the Auger spectrometer.

An additional advancement of significant importance is improvements in ion guns. This has been a two-pronged approach. Low-energy high-current ion guns provide improved depth resolution while minimizing the loss of chemical information from the sputtered layer. Combined, these permit more accurate identification of layer composition and thickness. The second approach is to incorporate FIB sources into the spectrometer chamber. FIB sources operate at very high current and very small beam diameters, allowing one to selectively ion mill stair-stepped trenches. This greatly reduces the time and effort required to expose buried particles or layers for subsequent AES analysis.

If past history predicts future advances, one should expect soon to see instruments that are faster<sup>21</sup> and offer greater ease of use for chemical analysis, not simply elemental analysis. Moreover, instruments tailored to specific industrial concerns will continue to be developed. This will most likely include industry-specific sample handling hardware as well as multi-tool analytical chambers.

## **8.3 X-RAY PHOTOELECTRON SPECTROSCOPY**

### **8.3.1 Background of XPS**

The evolution of XPS as a surface analytical method parallels AES in many ways. Both trace their foundation to turn-of-the-century studies of the properties of the atom. The photoelectric effect in which electrons are spontaneously



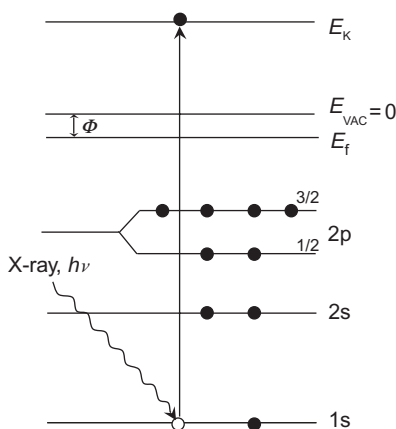
emitted from a material upon illumination by X-rays was discovered by Hertz in 1887.<sup>22</sup> Eighteen years later, Einstein described the basic equation of XPS that related the kinetic energy of the electron to the binding energy of the electron and the X-ray source energy.<sup>23</sup> In the 1920s the understanding of XPS progressed rapidly and it was shown that XPS spectra contained not only photoelectrons but also Auger electrons. It was not until the late 1940s that Steinhardt and Serfass envisioned XPS as an analytical tool for investigating surface chemistry.<sup>24</sup> Siegbahn and coworkers followed this lead and observed chemical shifts of core-level binding energies,<sup>25–28</sup> which was the basis of XPS for years to come. Siegbahn also coined the acronym ESCA to emphasize the fact that XPS spectra contained both photoelectrons and Auger electrons. To this day, the terms XPS and ESCA are used synonymously by practitioners even though this causes some confusion among lay users.

### 8.3.2 Basic Principles of XPS

XPS is based on the detection of electrons emitted from a surface under illumination with X-rays.<sup>5,29</sup> A schematic diagram of the photoelectron emission process is shown in Figure 8.7. If the X-ray source is mono-energetic, having an energy of  $h\nu$ , the kinetic energy,  $E_K$ , of the emitted electrons will be related to the binding energy of the electron  $E_B$  by the following equation:

$$E_K = h\nu - E_B - \Phi \quad (8.2)$$

In this equation  $\Phi$  is a small correction term used to account for the combined work function of the sample and the spectrometer. Note that electrons are emitted only if the X-ray source energy  $h\nu$  is greater than the electron binding energy and instrument work function,  $E_B + \Phi$ . Thus by measuring the kinetic energy of



**FIGURE 8.7** Schematic diagram of photoelectron emission. (Source: Evans Analytical Group, Sunnyvale, CA. Reprinted with permission.)

the emitted electron one can calculate the binding energy of that electron, that is, the energy needed to strip the electron from the atom. Electron binding energies are characteristic not only of the atoms from which the electrons originate but also of the chemical state or oxidation state of the atoms.<sup>30,31</sup> Except for hydrogen and helium, all elements in the periodic table are readily detected by XPS. Hydrogen and helium are undetectable due to their very low X-ray cross-sections. Thus XPS, at a minimum, is a qualitative elemental and chemical analysis method.

Electron binding energies are fundamental properties of the atom. Conversely, electron kinetic energies are not, since they vary with the X-ray source energy. Thus in XPS, electron energies are reported exclusively in terms of binding energies. An XPS spectrum is therefore a plot of the number of emitted electrons versus electron binding energy. One or more photoelectron peaks will be present in the spectrum for each type of atom in the sampling volume. Photoelectrons are generally classified as being either core electrons or valence electrons. The vast majority of photoelectrons are core electrons, which are used to determine elemental composition and chemical states (oxidation states). Valence electrons are of lower intensity but are extremely valuable for fingerprint identification of compounds.

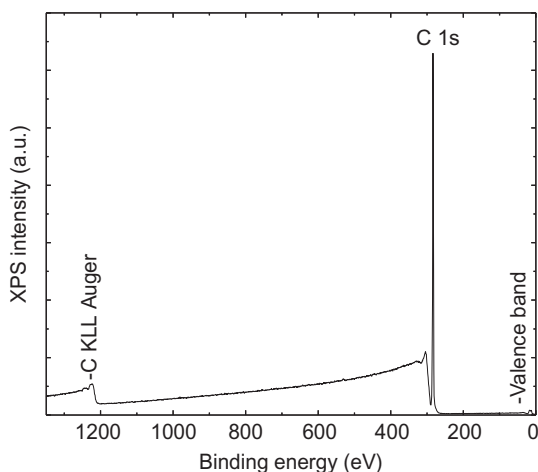
Photoemission of an electron from a core shell of energy  $E_x$  creates a core hole, thereby putting the atom in an excited state. Relaxation begins with the core vacancy being filled by an electron from an outer shell of energy  $E_y$ . The energy difference between the core and outer shells,  $E_x - E_y$ , is dissipated by photon emission, that is, X-ray fluorescence, or by electron emission from a shell of energy  $E_z$ , that is, Auger electron emission. The kinetic energy of the emitted Auger electron,  $E_{\text{Auger}}$ , is the transferred energy,  $E_x - E_y$ , minus the emitted electron's initial energy,  $E_z$ , that is

$$E_A = E_x - E_y - E_z \quad (8.3)$$

Because the energies of each shell are specific to the atom, the kinetic energy of the emitted Auger electron is also specific to the atom from which it originated. An XPS spectrometer cannot directly differentiate between photoelectrons and Auger electrons, and since XPS spectra are displayed in terms of binding energies, the same mathematical conversion is performed on both electrons, that is

$$E_B = h\nu - E_K - \Phi \quad (8.4)$$

even though the conversion has no meaning for Auger electrons. Since the kinetic energy of an Auger electron is a constant value dependent solely on the energies of the electrons involved in the Auger cascade, the “binding energy” of an Auger electron changes with X-ray source energy. Thus, a simple way to indirectly differentiate between photoelectrons and Auger electrons is to acquire spectra using two X-ray sources of different energies. If the binding energy of the peak remains constant, the peak is due to photoelectrons. If the



**FIGURE 8.8** XPS survey spectrum of polyethylene (PE). (Source: Evans Analytical Group, Sunnyvale, CA. Reprinted with permission.)

binding energy of the peak changes, the peak is due to Auger electrons. For a more detailed description of Auger electron generation, see [Section 8.2.2](#).

The very simple XPS spectrum of polyethylene,  $-\text{[CH}_2\text{]}_n-$ , a polymer containing only carbon and hydrogen, is shown in [Figure 8.8](#). Since hydrogen is not detected, the XPS spectrum consists of peaks due to carbon. A carbon atom has six electrons. The two 1s electrons in the K shell are considered to be core electrons and emission of these electrons creates the carbon 1s or “C 1s” peak. Since polyethylene contains only one type of carbon, a single C 1s peak is present. The two 2s and two 2p electrons in the L shell are considered to be valence electrons and generate a series of peaks between 0 and 25 eV. The shape of the valence band region is largely controlled by the band structure of the material. Since band structures are not easily estimated, the valence band region is often used as a fingerprint region with comparisons made to reference spectra from pure materials. Valence band spectra are particularly useful for differentiating organic compounds.<sup>32</sup> In addition to photoelectrons, the XPS spectrum contains carbon Auger electrons (C KLL) produced when L-shell electrons fill the K-shell vacancy.

As stated earlier, when a compound contains multiple elements one or more peaks will be present for each element, hence XPS can be used for qualitative analysis. Perhaps of more importance is the fact that photoelectron peak intensities are directly related to elemental concentration, hence XPS can be used for quantitative analysis. Theoretically, the intensity of the photoelectrons  $I_i$  is proportional to the X-ray flux  $f$ , the photoelectric cross-section  $\sigma_i$  for the corresponding electronic shell, the inelastic mean free path  $\lambda_i$  of the photoelectron, the analysis area  $A$ , the density  $n_i$  of atom  $i$  in the analysis volume, an angular asymmetry factor  $L$  based on the geometry of the spectrometer, the transmission

function of the spectrometer  $T$ , and the detection efficiency  $D$ .<sup>5</sup> Thus, the measured intensities  $I_i$  can, in principle, be related to atom density by

$$n_i \propto \frac{I}{f \sigma_i \lambda_i A L T D} \quad (8.5)$$

This method is not used because it is difficult to perform and has large experimental errors. In practice, an empirical method is used to calculate relative elemental concentration. By defining elemental sensitivity factors as  $S_i = f \sigma_i \lambda_i A L T D$ , the density equation (8.5) can be rewritten as

$$n_i \propto \frac{I_i}{S_i} \quad (8.6)$$

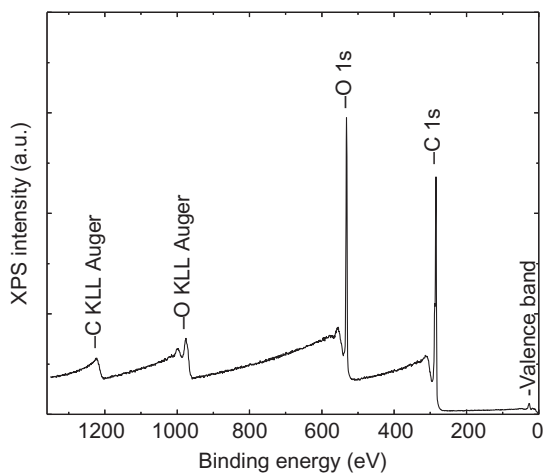
Note that  $S_i$  is different for each element and is dependent on instrument conditions, but is, to a first approximation, independent of chemical composition and state.

The general expression for calculating elemental concentrations expressed in atom percent is given by

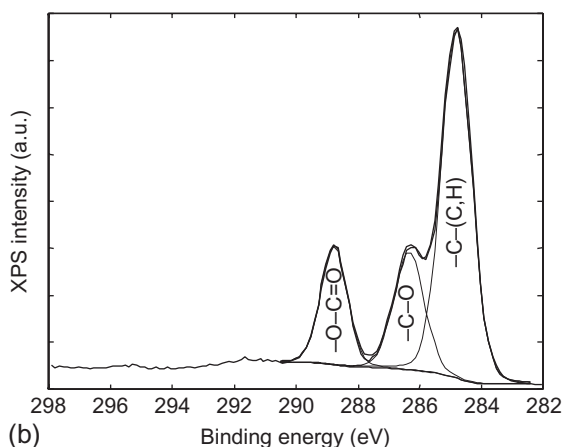
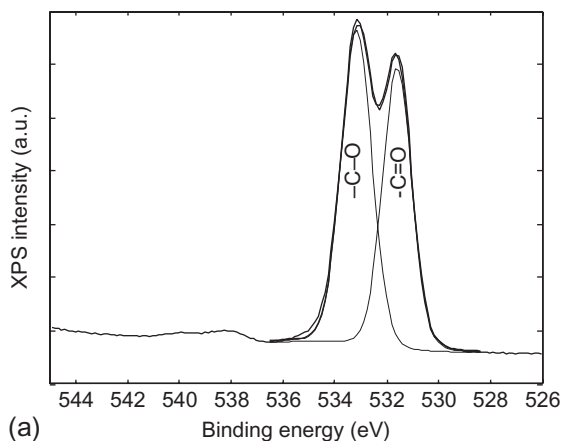
$$C_i = 100 \times \frac{I_i/S_i}{\sum_i I_i/S_i} \quad (8.7)$$

Because this equation uses intensity ratios one need not know the absolute magnitude of each peak, but only the relative peak intensities. Most commercial instrument manufacturers provide tables of sensitivity factors derived from pure compounds or photoelectron cross-sections, allowing straightforward calculation of relative elemental concentrations. Quantitative analysis is readily illustrated by examining the XPS spectrum of poly(ethylene terephthalate) (PET), a polymer containing two ester  $\text{O}-\text{C}=\text{O}$  functional groups  $[\text{O}(\text{O}=\text{C})\text{R}(\text{C}=\text{O})\text{OR}^1]_n$ , where  $\text{R}$  is  $\text{C}_6\text{H}_4$  and  $\text{R}^1$  is  $\text{C}_2\text{H}_4$ . Each monomer of PET contains 10 carbon, 4 oxygen, and 8 hydrogen atoms, giving rise to the survey spectrum shown in Figure 8.9. By measuring the intensities of one peak for each element, that is, the C 1s and O 1s peaks, and using the appropriate sensitivity factors, the atomic concentrations can be calculated from Equation (8.7). In this sample the analyzed volume contained 71.4 at.% carbon and 28.6 at.% oxygen. Because the sample was pure and the sensitivity factors were accurate, the measured concentrations exactly matched the theoretical values for this material.

Quantitative analysis of chemical bonds can also be achieved if the bonds generate distinct peaks in the spectrum. As shown in Figure 8.10(a), the oxygen 1s spectrum of PET contains two components,  $\text{O}-\text{C}$  at 533 eV and  $\text{O}=\text{C}$  at 532 eV. As shown in Figure 8.10(b), the carbon 1s spectrum contains three components,  $\text{C}-(\text{C}, \text{H})$  at 285 eV due to the aromatic ring,  $\text{C}-\text{O}$  at 286 eV and  $\text{O}-\text{C}=\text{O}$  at 288.5 eV. The intensity of each component is directly related to the concentration of atoms having the specified bond.<sup>32</sup> In this sample the two components of the oxygen spectrum have relative intensities of 1.1:1.0,



**FIGURE 8.9** XPS survey spectrum of poly(ethylene terephthalate) (PET). (Source: Evans Analytical Group, Sunnyvale, CA. Reprinted with permission.)

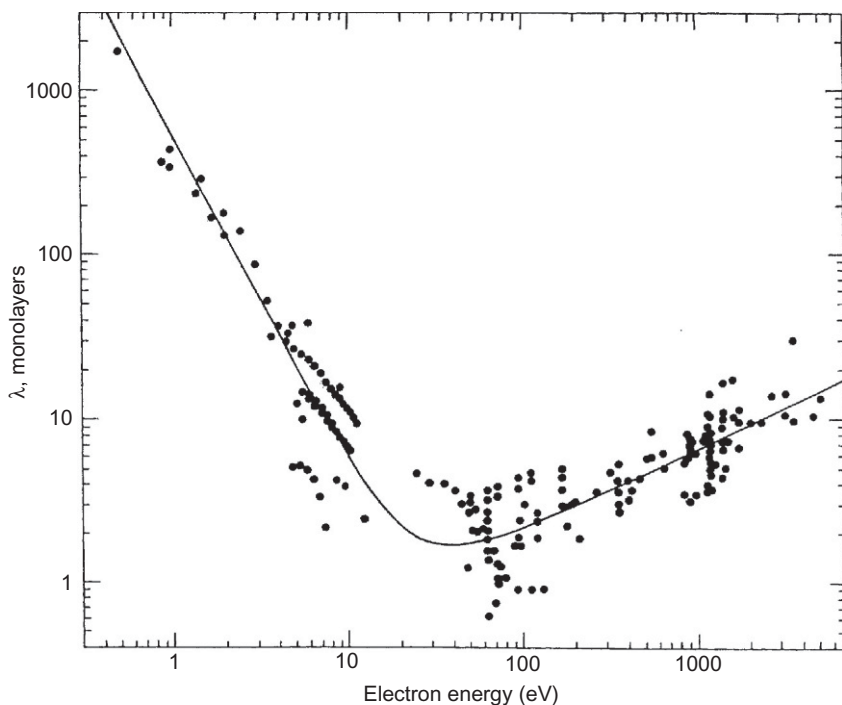


**FIGURE 8.10** High-resolution spectra of poly(ethylene terephthalate) (PET) showing the (a) oxygen 1s and (b) carbon 1s regions. (Source: Evans Analytical Group, Sunnyvale, CA. Reprinted with permission.)

in excellent agreement with the theoretical ratio of 1:1. Moreover, the three components of the carbon spectrum have relative intensities of 1.1:1.0:3.0, also in excellent agreement with the theoretical ratio of 1:1:3.

In XPS, detection of electrons requires that the electrons exit the material. As the electron travels through the solid toward the surface it can undergo elastic or inelastic scattering. Only those electrons that undergo elastic scattering exit the solid with their initial kinetic energy and hence contribute to the photoelectron peaks. Electrons that undergo inelastic scattering form energy loss tails and the background continuum. Inelastic collisions are random chance events and as such the inelastic mean free path,  $\lambda$ , is defined as the distance electrons travel after which 27% will have lost energy and 63% will still have their initial energy. For electrons with kinetic energies  $>50$  eV, this distance has been found to be roughly proportional to the square root of the kinetic energies of the electrons (see Figure 8.11). The probability that elastically scattered electrons can reach the surface and are detected decreases exponentially with distance,  $d$ , into the material and is given by

$$I_d = I^0 \left[ 1 - e^{-(d/\lambda \sin \theta)} \right] \quad (8.8)$$



**FIGURE 8.11** Inelastic mean free path  $\lambda$  versus emitted electron energy for elements. (Source: Seah, M.P.; Dench, W.A. *Surface Interface Anal.* 1, 2 (1979). Reprinted with permission.)



where  $\theta$  is the take-off angle, that is, the angle between the sample plane and the entrance to the electron analyzer. This means that there is no single XPS sampling depth since it depends on the probability of electron scattering, the electron take-off angle, and the inelastic mean free path, which varies with electron energy. For example, 63% of all electrons in a particular photoelectron peak originate from a depth of  $\lambda \sin \theta$ . Conversely, 95% and 99% of these electrons come from depths of  $3\lambda \sin \theta$  and  $5\lambda \sin \theta$ , respectively. Hence, the sampling depth can be based on detecting 63%, 95%, 99% or any portion of the elastically scattered electrons.

It is important to note that sampling depth can be varied by changing the take-off angle,  $\theta$ . At a take-off angle of  $90^\circ$ , the depth from which 63% of all detected electrons originate is  $\lambda \sin(90^\circ)$  or  $\lambda$ . At a grazing take-off angle of  $10^\circ$ , the sampling depth drops to  $\lambda \sin(10^\circ)$  or  $0.17\lambda$ . For carbon electrons in polymers,  $\lambda$  is  $\sim 3$  nm, hence the 63% sampling depths are 3 nm and 0.5 nm at take-off angles of  $90^\circ$  and  $10^\circ$ , respectively. This makes XPS an extremely surface-sensitive technique. By changing the take-off angle, and consequently the sampling depth, one can obtain non-destructive depth profiles from the outermost surface to  $\sim 10$  nm.<sup>33</sup> Moreover, if the contamination layer is thin ( $< 10$  nm), one can also estimate the thickness.<sup>34</sup> When contaminant layers are thicker than 10 nm, non-destructive profiling is not possible. In these cases destructive depth profiling methods are needed. Destructive depth profiling can be performed by a variety of mechanical and chemical means. The most popular method is ion beam etching, wherein the surface is first etched with an ion beam, such as  $\text{Ar}^+$ , and the freshly exposed surface is then analyzed. This sequence of steps is repeated until the desired sputter depth is reached. A depth profile plot thus presents the intensity or concentration of the measured peak versus sputter time, or if the sputter rate is known, versus sputter depth.

The consequences of ion beam sputtering warrant further discussion to prevent erroneous interpretation of profile data. When a high-energy incident ion strikes a surface, the energy of the incident ion is dissipated by the surface atoms, resulting in bond breaking and ejection of material from the surface. For typical ion beam voltages of  $< 5$  kV, the damaged layer is a few nanometers thick. For some materials such as silicon and aluminum oxides the damaged layer has the same elemental and chemical composition as the undamaged material. Unfortunately, this is often not the case and ion sputtering results in two artifacts: preferential sputtering and ion beam damage. As the name implies, preferential sputtering results in elements being sputtered at different rates. This causes a change in the elemental composition within the damaged layer. For example, when  $\text{Ta}_2\text{O}_5$  was ion sputtered using 3 keV  $\text{Ar}^+$  ions, the Ta:O ratio changed from 2:5 ( $\text{Ta}_2\text{O}_5$ ) to 2.3:1 ( $\text{Ta}_{2.3}\text{O}$ ). Since oxygen is removed at a faster rate than tantalum, tantalum sub-oxides are formed. This classic example illustrates that when ion sputtering is performed one cannot rely on the intensity and location of XPS peaks to be representative of the unsputtered material. Rather, the extent of preferential sputtering must be investigated

for each material. From these control studies one can calculate sensitivity factors appropriate for sputtered materials. Although one can often correct for the effect of preferential sputtering on peak intensities, the effects of ion beam damage are more problematic. Even in the absence of preferential sputtering, ion beam damage causes bond breaking and consequential creation of new bonds. Thus, chemical bonding information is often lost. For example, ion sputtering of PET results in preferential sputtering of oxygen. Consequently, the remaining carbon atoms recombine to form a material resembling an organic char. Unlike with preferential sputtering, once chemical information is lost it cannot be retrieved. Thus, when chemical information from the sub-surface region is required one must minimize the thickness of the ion beam damaged layer. To some extent this can be accomplished by lowering the ion beam energy and the incident angle, and by using polyatomic ion beams<sup>35</sup> such as  $\text{SF}_5^+$  or  $\text{C}_{60}^+$ .

All of the examples presented about have demonstrated the use of XPS for organic compound analysis. This is rightly so since XPS is one of the few techniques that provides organic compound identification. However, XPS is also widely used for analysis of inorganic materials. Since changes in the chemical environments of atoms cause a shift in binding energy of the electrons, one can identify different oxidation states. In general, the binding energies of the core electrons increase with increasing oxidation of the element. For most metals XPS can easily distinguish between the metallic and oxide states. For example, the titanium  $2p^{3/2}$  peak in metallic titanium is found at 453.8 eV whereas in titanium dioxide it is at 458.5 eV. In some instances XPS can even distinguish between different oxidation states. For example the  $3d^{5/2}$  peak of niobium is at 202 eV for Nb, 204 eV for NbO, 206 eV for NbO<sub>2</sub>, and 207.6 eV for Nb<sub>2</sub>O<sub>5</sub>. However, for some elements chemical shifts due to differences in oxidation states are of the same magnitude as the shifts due to differences in chemical states. For example, the  $2p^{3/2}$  binding energies of iron as  $\text{Fe}^{2+}$  and  $\text{Fe}^{3+}$  are 709.6 and 711 eV in FeO and Fe<sub>2</sub>O<sub>3</sub>, but are 710.6 and 711.3 eV for FeCl<sub>2</sub> and FeCl<sub>3</sub>. For electronegative elements the binding energies of the core electrons decrease with increasing reduction of the element. For instance, reduction of  $\text{S}^0$  to  $\text{S}^{2-}$  causes a decrease in the electron binding energy from 164.1 to 161–162 eV. This downward shift upon reduction is observed for fluorine, chlorine, bromine, iodine, oxygen, phosphorus, sulfur, arsenic, and selenium.

Occasionally spectra contain loss features such as plasmons, phonons, and  $\pi \rightarrow \pi^*$  peaks or relaxation-related features such as shake-up and shake-off peaks.<sup>5</sup> These peaks appear on the high-binding-energy side of the core level peak and can provide clues to chemical states. Although the origins of these peaks are complex and not easily predicted, their presence or absence can be used to identify oxidation states of many elements such as aluminum, silicon, magnesium, cobalt, nickel, copper, lanthanum, and cerium. Take, for instance, when oxides or stains are formed on metallic copper, the existence of strong shake-up peaks indicates the formation of  $\text{Cu}^{2+}$  (e.g., CuO and CuSO<sub>4</sub>). While the lack of shake-up features implies that the surface is  $\text{Cu}^0$  or  $\text{Cu}^{1+}$  (e.g., Cu<sub>2</sub>O).



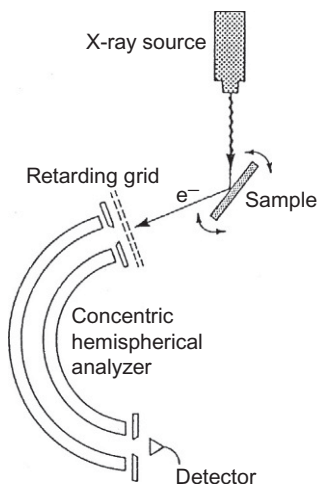
For some elements chemical shifts are small and satellites features are absent. In these cases the only means left for deducing chemical information is by examining the position of Auger peaks.<sup>36,37</sup> This is most often done by calculating the “Auger parameter,”  $\alpha$  (actually the modified Auger parameter),

$$\alpha = E_B(i) + E_K(jkl) \quad (8.9)$$

where  $E_B(i)$  is the binding energy of the photoelectron  $i$  and  $E_K(jkl)$  is the kinetic energy of the Auger electron  $jkl$ . The Auger parameter is often used to identify chemical composition and oxidation states in fluorine, sodium, silicon, copper, zinc, arsenic, silver, cadmium, indium, tellurium, and lead. For example, the binding energies of  $\text{Cu}^0$  and  $\text{Cu}^{1+}$  are nearly the same,  $\sim 932.5$  eV. However, these two species are readily distinguished since their Auger parameters are 1851.2 and 1849.4 eV, respectively.

### 8.3.3 XPS Instrumentation

The typical components in an XPS system include an ultra-high-vacuum (UHV) chamber, one or more X-ray sources, an energy analyzer, a multi-axis sample stage, an imaging device, a sample neutralizer, an ion gun, and one or more computers for data acquisition, storage, and manipulation. A simplified schematic of an XPS system is shown in Figure 8.12. Common polychromatic X-ray sources generate aluminum X-rays, for example, the  $\text{Al } K\alpha_{1,2}$  doublet of energy 1486.6 eV with a 0.85-eV line width, or magnesium X-rays, for example, the  $\text{Mg } K\alpha_{1,2}$  doublet of energy 1253.6 eV with a 0.7-eV line width. To improve spectral resolution a monochromator is often used to select the Al



**FIGURE 8.12** Schematic diagram of an XPS system. (Source: Evans Analytical Group, Sunnyvale, CA. Reprinted with permission.)

$K\alpha_1$  singlet, thereby reducing the source width to 0.35 eV. Moreover, monochromatic X-ray sources greatly reduce or eliminate damage caused by heat, electrons, and Bremsstrahlung radiation generated by polychromatic X-ray sources. The most common analyzer is a concentric hemispherical analyzer (CHA). This analyzer has good absolute energy resolution and can be operated at a fixed energy resolution independent of electron energy. A multi-axis sample stage facilitates sample positioning and allows analysis at different sampling depths by varying the electron take-off angle. Cooling and heating stages are useful for preservation of volatile components and thermal sample modifications, respectively. Specific analysis areas are routinely determined from optical and X-ray-induced secondary electron images or from XPS elemental and chemical maps. Unless specimen neutralizers are used, insulating samples can develop positive surface charges during XPS analysis, which cause distortion and shifting of peaks. Thus, most spectrometers are equipped with sample neutralizers such as electron flood guns, or low-energy ion guns, or polychromatic X-ray sources.

Most XPS spectrometers can acquire a variety of data types. The most common are survey spectra, high-resolution spectra, and depth profiles. In these modes the analysis area is generally dictated by the X-ray source illumination area or by transfer lenses and apertures. Survey spectra are acquired in a mode whereby energy resolution is sacrificed in order to improve detection limits. Moreover, as the name implies, data are acquired over all binding energies so as to survey all elements. Quantitative precision is generally lower when data are acquired in survey mode. Conversely, high-resolution spectra are acquired in a mode whereby detection limit is sacrificed in order to improve energy resolution. High-resolution spectra typically encompass specific energy windows of interest rather than all energies. These spectra are used for high-precision quantitative analysis and chemical bonding studies. Depth profiling is typically performed by interlacing ion beam sputtering with high-resolution data acquisition.

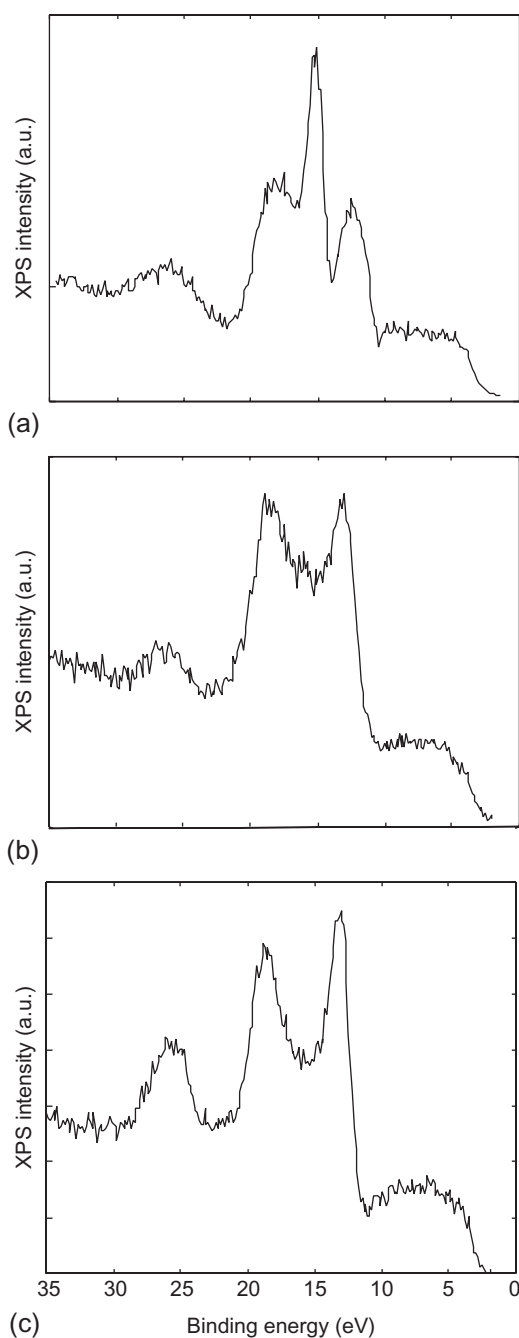
Elemental and chemical mapping are available on current commercial instruments. Mapping is performed by four methods: sample translation, rastering the transfer lens, rastering the X-ray source, or direct imaging onto a multi-channel detector. In each mode the result is a pseudo-color image showing the intensity of a selected peak as a function of  $X$ ,  $Y$  location on the surface. Direct imaging has the advantage of rapid image acquisition, whereas transfer lens and X-ray source rastering assures that subsequent selected area analysis is in perfect spatial alignment with the imaged area. Sample translation is the slowest method but permits mapping of very large areas.

### 8.3.4 Applications of XPS for Characterizing Surface Contaminants

The need to monitor and control contaminants spans almost every manufacturing industry. XPS has been widely used for cleaning and contamination studies

in diverse industries such as aerospace, automotive, biomedical, chemical, catalysis, electronic equipment, medical device, packaging, paint, paper, pharmaceutical, plastics, polymer, printing, and semiconductor. Surface contaminants can be categorized into two major groups, organic and inorganic materials. Common organic contaminants include lubricants, oils, waxes, greases, mold-release agents, antioxidants, plasticizers, surfactants, and adventitious airborne compounds. Common inorganic contaminants include metals, compounds, adsorbed cations or anions, corrosion products, and oxides. When one considers the number of common contaminants along with the number of potential substrates from just the few industries highlighted above, it becomes obvious that a detailed comprehensive review is impossible. Examples included in the following discussions highlight routine uses of XPS to study (1) surface composition, (2) adhesion failure, (3) corrosion, and (4) cleaning methods. XPS is uniquely suited to these studies because it can examine contaminants that are solids or low-volatility liquids on both conductive and insulating substrates. Within these bounds the primary limitations of the method are in detection limits and minimum analysis area. The minimum detectable level for most elements is a fraction of a monolayer or a bulk concentration of 0.05–0.5 at.%. The current state-of-the-art minimum analysis area is 10  $\mu\text{m}$  in diameter.

It should be no surprise that synthetic materials are often contaminated since contamination can take place at all phases of a product's life cycle. Polymers are generally either very clean or they contain significant concentrations of surface contaminants. Most of these "contaminants" are either foreign compounds found only at the surface or additives that have migrated to and concentrated at the surface. While the latter might not fit a strict definition of a contaminant, they are included here since they often have similar detrimental effects on product performance. Many compounds can be identified by elemental and chemical analysis using core electron spectra. For example, by examining the positions and intensities of the silicon 2p and oxygen 1s peaks it is easy to detect and quantify poly(dimethylsiloxane) (PDMS),  $-\text{O}-\text{Si}(\text{CH}_3)_2\text{--}$ , contamination on polyolefin film surfaces. Conversely, polyolefin contamination by fatty acid amides such as behenamide,  $\text{H}(\text{CH}_2)_{21}\text{CONH}_2$ , or erucamide,  $\text{H}(\text{CH}_2)_8\text{CH}=\text{CH}(\text{CH}_2)_{11}\text{CONH}_2$ , common slip agents added to aid processing, is more difficult to identify since most of the atoms of these compounds are part of the olefin chain. Using the core electron spectra, only the amide end group,  $\text{CONH}_2$ , distinguishes it from the substrate. However, if the polyolefin is polypropylene, then the valence band spectrum can be used for identification since the pendent methyl groups of polypropylene produce a valence band spectrum distinctly different from the unbranched tail of a fatty acid amide. For example, upon heating an oriented polypropylene (OPP) film that contained myristamide,  $\text{H}(\text{CH}_2)_{13}\text{CONH}_2$ , 4 h at 65  $^\circ\text{C}$ , the fatty acid amide bloomed to the surface. Quite different from the valence band spectrum produced by the branched carbon of the as-received OPP film [see Figure 8.13(a)], the valence band spectrum of the heated OPP film [Figure 8.13(b)] resembled



**FIGURE 8.13** Valence band spectra of oriented polypropylene containing myristamide: (a) as-received and (b) after heating for 4 h at 65 °C, and (c) of pure myristamide. (Source: Evans Analytical Group, Sunnyvale, CA. Reprinted with permission.)

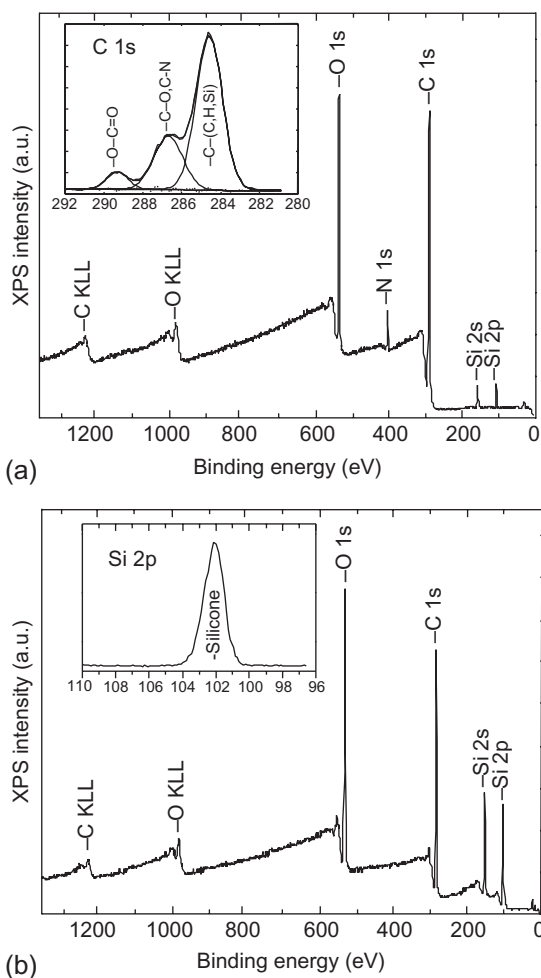
that produced by the unbranched tail of pure myristamide [Figure 8.13(c)]. In fact, valence band spectra are useful in determining the structure of many additives.<sup>38</sup>

In many ways assessing contaminants on inorganic materials is easier since the contaminants often contain foreign elements not found in the pure material. Since XPS can detect all elements except hydrogen and helium, it is well suited for these studies. Most inorganic compounds have high surface free energies and become coated with adventitious airborne organic compounds. Thus, carbon is almost always detected on inorganic materials. Moreover, exposure to air and moisture often leads to surface oxidation. For instance, two commercially available high-purity silver halide powders, AgF and AgF<sub>2</sub>, were examined by XPS.<sup>39</sup> Even after brief sputtering with 1 keV He<sup>+</sup> to reduce surface contaminants, the surfaces still contained carbon and oxygen in various organic forms, silver oxides, and perhaps even CHF- and CFO(OH)-like species. Clearly these surfaces were not pure AgF and AgF<sub>2</sub>. This illustrates the general principle that one should never simply assume that surface and bulk compositions of inorganic compounds are the same.

There are times when a particular processing step is needed; however, the method by which it is accomplished drastically changes surface composition. For example, orthopedic implants must be sterilized prior to insertion into the body. While the efficacy of autoclaving for killing pathogens is known, it is less well known that the method often deposits surface contaminants. It has been shown that steam-autoclaved titanium surfaces contain high levels of oxygen- and nitrogen-containing organic compounds and a variety of inorganic contaminants.<sup>40</sup> Conversely, ultraviolet or  $\gamma$  irradiation produces much cleaner surfaces.<sup>40</sup> In some materials, surfaces initially appear clean only to have contamination develop with time. For instance, migration of bulk contaminants along grain boundaries of electroplated metals can take place, particularly if the sample experiences elevated temperatures. This phenomenon was reported recently for electrolytically gilded silver.<sup>41</sup> After 8 months of indoor storage, red spots of gold chloride were observed. Contamination also can take place during use of a product. This is quite true for catalytic materials such as automotive exhaust converters. In one simulation study, contaminants such as phosphorus, lead, zinc, and calcium were found to deactivate the catalyst.<sup>42</sup> The main sources of these contaminants were believed to be lubricant additives.

XPS has been used to investigate adhesion failure for many years.<sup>43</sup> Studies have found that many adhesion failures are caused by contaminants at interfaces. Among all the materials that can cause adhesion failure, PDMS is probably the most ubiquitous, being used as mold release agents, lubricants, surfactants, anti-foaming agents, and plasticizers. There are literally thousands of materials containing PDMS and it is not easy to avoid exposure to them. Unfortunately, low-molecular-weight fractions can cover bonding surfaces, either through direct contact with a PDMS source or through surface segregation and migration. For example, PDMS mold release agents can concentrate in molds if too much release

is used or if the molds are improperly cleaned. Parts produced in such molds can have silicone layers from a few nanometers to several tens of nanometers thick. Attempts to join these components either through adhesive or thermal bonding often result in weak adhesive joints or total adhesion failure. In one case where molded polypropylene was bonded to steel with an ethyl cyanoacrylate adhesive, complete adhesion failure was observed. As shown in [Figure 8.14\(a\)](#), the XPS spectrum taken from the steel side of the failure was a mixture of adhesive and silicone, whereas the spectrum taken from the polypropylene side of the failure was nearly pure silicone [[Figure 8.14\(b\)](#)]. Even seemingly clean

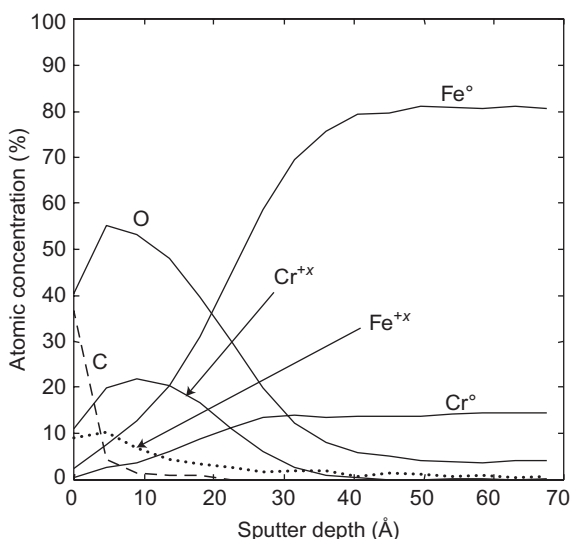


**FIGURE 8.14** XPS spectra from: (a) the steel side and (b) the polypropylene side of a failed polypropylene-ethylcyanoacrylate-steel adhesive joint. (Source: Evans Analytical Group, Sunnyvale, CA. Reprinted with permission.)

environments can, with time, become contaminated with PDMS even though no source is apparent. For example, it was observed that a cleanroom table on which the fiber-reinforced composite plies were handled was contaminated with PDMS.<sup>44</sup> Transfer of PDMS from the contaminated tabletop to the composite plies resulted in poor composite adhesion. Silicones can even come from products designed to prevent contamination such as disposable gloves.<sup>45</sup> The surfaces of disposable vinyl gloves are quite often covered with plasticizers such as PDMS. Although these gloves are used ostensibly to prevent sample contamination, they can in fact act as a source.

XPS is equally useful for investigating adhesion failure caused by other organic and polymeric materials. Low-molecular-weight fluorocarbons were observed to migrate in a plasma reactor used to deposit multilayer films for corrosion protection of aluminum alloys.<sup>46</sup> In this process trimethylsilane (TMS) and hexafluoroethane were sequentially deposited by a cathodic DC plasma. It was found that this process produced weakly adhered films. Poor adhesion was traced to fluorocarbon migration from the chamber walls to the sample during evacuation of the chamber. However, if an oxygen plasma cleaning step was introduced to clean the alloy before deposition of the TMS film, strongly adherent films were produced. Unfortunately, this process converted the fluorocarbon layer to a fluoride-containing oxide layer which was considered detrimental to corrosion resistance. Another typical example where adhesion failure was traced to a prior processing step was the failure of gold-plated nickel contacts in ceramic packages.<sup>47</sup> In this study a 20-nm-thick carbon layer was found on a freshly failed contact. The XPS carbon 1s spectrum obtained from the contact matched that of a wax used in a process step prior to plating. Since the root cause of failure was wax buildup, adhesion was restored by more frequent changing of the cleaning solutions.

XPS is often used to study corrosion since the technique provides elemental and chemical information. Many metals and alloys have passive oxide layers that limit corrosion. For example, bronzes were found to have a protective surface layer of lead oxide. However, when this layer was destroyed, such as by exposure to a moist  $\text{SO}_2$  environment, copper oxide and copper sulfite were formed.<sup>48</sup> Similarly when copper nickel alloys were exposed to synthetic sweat, for example, to mimic contact of earrings, jewelry and watches with skin, copper(I) oxide and chloride were formed on high-copper alloys.<sup>49</sup> Conversely, on high-nickel alloys, copper(II) hydroxychloride, nickel hydroxide, and nickel oxide were formed. Corrosion of ferrous alloys has also been extensively analyzed, once again due to the ability of XPS to determine chemical states. Moreover, depth profiling by ion beam sputtering was often used during these experiments to measure the in-depth composition and the thickness of the corrosion layers. In one study of iron-aluminum intermetallics exposed to traces of  $\text{H}_2\text{S}$  at 600 °C, the scale formed consisted of three layers: an outermost organic layer, a mixed layer of  $\text{FeS}$  and  $\text{Al}_2\text{O}_3$ , and a bottom layer of  $\text{Al}_2\text{O}_3$ .<sup>50</sup> In another study, chemical state depth profiling was obtained for an electropolished



**FIGURE 8.15** Chemical depth profile of an electropolished stainless steel sample. Note that sputter depth is based on the sputter rate of  $\text{SiO}_2$ . (Source: Evans Analytical Group, Sunnyvale, CA. Reprinted with permission.)

stainless steel surface (see Figure 8.15). The outermost surface contained an organic layer that was less than a monolayer thick. Both  $\text{Cr}^{x+}$  and  $\text{Fe}^{x+}$  oxides were detected, with  $\text{Fe}^{x+}$  being concentrated at the outer surface. The approximate thicknesses of the  $\text{Cr}^{x+}$  and  $\text{Fe}^{x+}$  layers were 2 and 0.9 nm, respectively, assuming that this material sputtered at the same rate as  $\text{SiO}_2$ . Corrosion not only results in cosmetic blemishes and surface damage but can cause poor adhesion. This is particularly important in the microelectronics industry. For example, good adhesion of gold-ball bonds to bond pads is vital since it is the first external link allowing information to travel from the IC device to board-level components. In one study ball-bond failure was traced to a reactive ion etching step performed after the pad was formed.<sup>51</sup> Since bond-pad production is not the last step in device fabrication, one must bear in mind the effect of subsequent processing steps on bond-pad cleanliness.

Although production methods can be selected to minimize surface contamination, often the resulting surface requires additional cleaning processes. Because the matrix of contaminants, cleaning methods, and substrates can be large, it is imperative to examine the efficacy of the method chosen for a given substrate. For many systems this is performed easily by XPS. Not only does the method permit quantitative evaluation of contaminant removal, but it can also verify that the substrate composition has not been altered and that new contaminants were not introduced during the cleaning process. A gentle method for removing organic compounds from inorganic surfaces is simply to heat the material. For example, heating NiP/Al or hard-disk glass substrates to 400 °C



removes nearly all of the carbon contamination.<sup>52</sup> Care must be taken, however, to assure the substrate is not damaged or modified by thermal exposure. In fact, surface segregation of mobile elements took place at temperatures as low as 300 and 350 °C for the glass and NiP/Al substrates, respectively. Thus, only a portion of the organic contaminants could be removed without sample damage.

Wet chemical cleaning methods have also been studied by XPS. When stripping photoresist from aluminum, silicon, and silicon dioxide surfaces it is desirable to remove all traces of photoresist while leaving the substrates unaffected. Low-molecular-weight alcohols, for example, isopropanol, were demonstrated to remove photoresist and produced surfaces with organic contamination levels similar to the RCA cleaning method.<sup>53</sup> The process appears, however, to leave a residual adhesion promoter layer that was applied prior to the photoresist. Dry cleaning methods such as oxygen-containing plasmas are well known to remove organic residues from surfaces<sup>54</sup> but the cleaned surfaces are typically oxidized.<sup>55</sup> While this would not be a problem for silicon dioxide surfaces, the growth of oxides on aluminum substrates such as bond pads is undesirable since it impairs wire bonding. Plasmas such as Ar/H<sub>2</sub> have been shown to remove the surface oxide from bond pads but the process may also damage adjacent passivation layers.<sup>19</sup> Clearly the conditions under which plasma cleaning is performed must be carefully monitored and controlled.

The cleaning examples presented above demonstrate that it is difficult to select a cleaning method that is both effective and harmless to the substrate. However, that is not to say that it cannot be done. In the case of indium tin oxide (ITO) substrates used in organic light-emitting diodes (OLEDs), the goal was to remove organic compounds that raised the turn-on voltage and lowered brightness. For this system, cleaning under ultraviolet light in conjunction with ozone gas, that is, UV/O<sub>3</sub> treatment, was found to significantly reduce organic contamination<sup>56,57</sup> while having no apparent effect on the chemical composition of the ITO substrate.<sup>56</sup>

### 8.3.5 Recent Developments and Future Directions of XPS

The four most important areas in which XPS has progressed over the past few years include analysis speed, minimum analysis area, imaging, and depth profiling. Analysis speed and sample throughput continue to climb due to advances in electronics, multi-channel detectors, computers, and software. Improvements in electronics and detectors have increased data acquisition rates. In addition, advances in software have allowed rapid identification and quantification of surface elements, often with little or no user intervention. Moreover, computer-controlled automation permits rapid sample positioning for both routine and complex multi-sample analyses. Large-sample platens are often used to facilitate automated analysis of multiple samples or multiple areas on a large sample. The largest platen that is commercially available is designed specifically to hold 300-mm-diameter silicon wafers.

Small-area XPS analysis is one of the major areas that has experienced improvements in recent years. Two approaches are available for decreasing analysis areas: use of microprobe<sup>58</sup> X-ray sources, and a combination of apertures and input lenses.<sup>59</sup> The microprobe method uses a focused electron beam that strikes the X-ray anode surface to generate X-rays. The X-rays are re-focused onto the analysis area by a crystal monochromator. The most recent microprobe system has a minimum analysis area of 10  $\mu\text{m}$  diameter. The aperture method uses a broad X-ray source to generate photoelectrons from a large area of the sample. Apertures and input lenses are used to define the analysis area. With this scheme, the smallest area that has been achieved is 12–15  $\mu\text{m}$  diameter.

XPS imaging is another area that has advanced significantly in the last few years. The two methods presently used for producing XPS images or maps are microprobe imaging<sup>60</sup> and parallel imaging.<sup>59,61</sup> In microprobe imaging, images are generated either by scanning a focused X-ray probe beam across the analysis area, or moving the analysis area under the probe beam. The spatial resolution of microprobe imaging is defined by the X-ray spot size, and is of the order of  $\sim 10 \mu\text{m}$ . In parallel imaging, either a polychromatic or a monochromatic X-ray beam is used. Images are simultaneously collected from the entire analysis area utilizing a two-dimensional array detector. The best spatial resolution in this mode is currently 2–3  $\mu\text{m}$ .

Finally, the need for high depth resolution and fast depth profiling has spurred advances in ion-gun design. The introduction of low-energy, high-current ion guns provides improved depth resolution during profiling while minimizing artifacts from the sputter damaged layer. Moreover, fine-resolution FIBs are now available that can be used to sputter small areas or to reduce the sputtering time when depth profiling.

Given the vast number of industries in which XPS analysis is employed it is not surprising that instrument manufacturers are being pulled in many directions. Fortunately, many of the latest improvements are applicable to a large majority of users. One can expect future instruments to offer greater ease of use, smaller analysis areas, faster acquisition times, and chemical depth profiling.

## 8.4 TIME-OF-FLIGHT SECONDARY ION MASS SPECTROMETRY

### 8.4.1 Background of TOF-SIMS

TOF-SIMS is one of the few techniques that can provide specific compound identification of molecules on a surface. Conceptually TOF-SIMS is rather simple: bombardment of a surface by high-energy ions causes emission of secondary ions, which are mass-analyzed by measuring the time required for the ions to fly from the sample to the detector. The difficulty, however, is in the details, that is, producing a well-defined short-duration primary ion pulse and accurately

measuring the secondary ion flight time. Much of the early development of TOF-SIMS was carried out by Professor Benninghoven and his team at the University of Münster in Germany.<sup>62</sup> It was not until the early 1980s that TOF-SIMS was developed into a viable analytical method. Although acceptance of the technique was initially slow, it has evolved into a standard tool for identifying organic and inorganic contaminants on surfaces. The compound specificity of TOF-SIMS clearly differentiates it from other chemically sensitive surface analysis methods such as XPS, also known as ESCA.

### 8.4.2 Basic Principles of TOF-SIMS

The generic class of SIMS methods is based on measuring charged species emitted from a surface due to bombardment by high-energy ions. Specifically, when a primary ion strikes the surface of a solid its kinetic energy is dissipated by the solid through collisional cascades within the near-surface region. The end result is that the primary ion is buried below the surface, bonds near the impact site are broken, and atoms, molecular fragments, and molecules are ejected from the top one to three atomic layers. Most of the ejected species have no net charge. However, a small fraction of these species will be either positively or negatively charged. Only these charged species (ions) are detected by the mass spectrometer, hence the term “SIMS.” If the primary ion dose is below  $\sim 10^{12}$  ions/cm<sup>2</sup>, the probability that two primary ions will strike the same location (on the atomic scale) is low and the measurement is said to have been performed in the static mode. That is, the surface is unchanged, or static during the measurement. Thus nearly all ions emitted from the sample originate from virgin, unsputtered material, and hence are characteristic of the original surface. If the primary ion dose greatly exceeds  $10^{12}$  ions/cm<sup>2</sup>, there is a significant probability that the emitted ions will arise from a previously sputtered area, and thus are not representative of the virgin surface. In this case the measurement is said to have been performed in the dynamic mode. Dynamic SIMS provides elemental composition typically as a function of sputter depth. Unfortunately, it does not provide the chemical information, as does static SIMS. For a review of dynamic SIMS (see Ref. 63).

Dynamic SIMS is preferred when one needs to determine the in-depth concentration of contaminants. Remember, however, that dynamic SIMS provides only elemental analysis. Conversely, static SIMS is usually preferred for identification of surface contaminants. This is particularly true when chemical information is needed, such as for organic compound identification. There are three types of static SIMS instruments differing in the mode of ion separation and detection: magnetic-sector, quadrupole, and time-of-flight spectrometers. Magnetic-sector spectrometers can provide excellent detection limits and high mass resolution. They are limited, however, in mass range, generally  $<300$  amu, and scan speed. Quadrupole spectrometers have the advantage of high scan speed, mass range up to 1000 amu, ease of use, and adaptability to hybrid systems. The primary

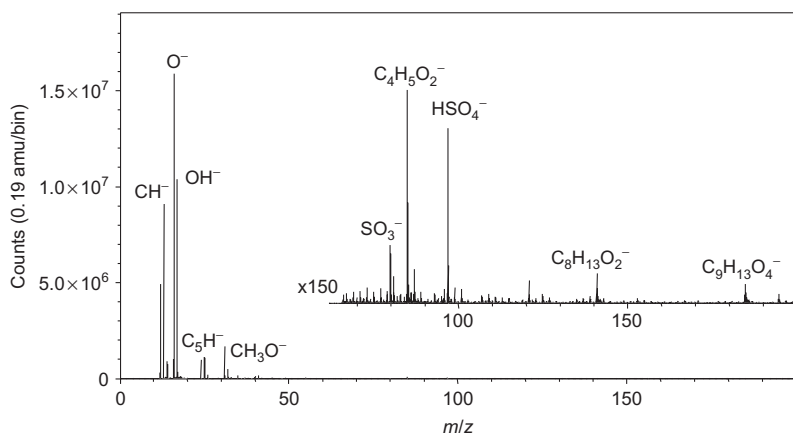
limitation of quadrupole systems is poor mass resolution, resulting in the inability to distinguish between ions of the same nominal mass. Time-of-flight spectrometers have high mass range (up to several thousand amu), high mass resolution ( $> 10,000$  m/ $\Delta m$  above 100 amu), and rapid data acquisition. The main disadvantage of TOF-SIMS is the low current of the primary ion beam, resulting in very slow depth profiling. In general, TOF-SIMS is the preferred method for static SIMS.

In a time-of-flight spectrometer ions of charge  $q$ , typically  $\pm 1$ , are subjected to an extraction field,  $V_e$ , of several thousand volts, imparting each ion with a fixed kinetic energy,  $E_k$ . Kinetic energy is given by  $E_k = qV_e$ , which is also equal to  $1/2mv^2$ , where  $m$  is the mass of the ion and  $v$  is the velocity of the ion. Since  $E_k$  is constant for all ions, the velocity of each ion is inversely proportional to its mass. Lighter ions reach higher velocities in the extraction field. Hence for a given flight path length,  $l$ , the flight time,  $t$ , is shorter for lighter ions,  $t = l/v$ . Heavier ions have lower velocities and travel more slowly. Thus, the mass of any ion having unit charge can be determined from the flight time,

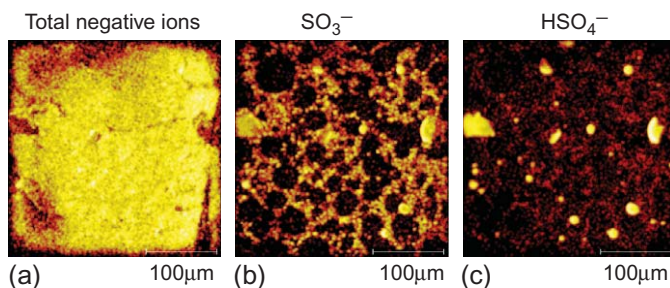
$$m = 2qV_e t^2 / l^2 \quad (8.10)$$

In practice  $V_e$ ,  $t$ , and  $l$  are not explicitly known so mass spectra are internally calibrated from ions of known masses, such as  $\text{CH}_3^+$ ,  $\text{C}_2\text{H}_3^+$ ,  $\text{OH}^-$ ,  $\text{C}_2\text{H}^-$ , etc. An important feature of TOF-SIMS is that a complete mass spectrum is obtained from each primary ion pulse. Moreover, a higher mass resolution can be obtained by using longer flight tubes. This description has omitted two important items, namely that the primary ions must be produced in packets and that the generation time of each packet should be as short as possible. This is necessary because the spread in generation time,  $\Delta t$ , directly affects mass resolution,  $(\Delta m)^{1/2}$ .

The spectrometer described above generates a plot of ion counts versus mass/charge. Since the vast majority of all ions have a charge of  $\pm 1$ , mass/charge and mass are almost synonymous terms. Figure 8.16 shows a typical mass spectrum. This figure shows the negative ion spectrum of a mixture of the antibiotic Tobramycin sulfate and Simplex-P bone cement, which itself is a mixture of poly(methyl methacrylate) (PMMA) and  $\text{BaSO}_4$ . Note that ions can be positively or negatively charged but only one polarity is present in a given spectrum. Thus for complete characterization of a material one must obtain both positive ion and negative ion spectra. In addition to spectral analysis, TOF-SIMS is routinely used for ion imaging. Ion images or maps are two-dimensional pseudo-color images showing the intensity of a given emitted ion, or sum of ions, as a function of  $X$ ,  $Y$  location on the surface. Figure 8.17 shows the distribution of PMMA,  $\text{BaSO}_4$ , and Tobramycin sulfate in a cross-sectioned sample of a bone cement/antibiotic mixture. This specimen was taken from residual material in an injection nozzle used during a hip replacement surgery. The ions used to image PMMA,  $\text{BaSO}_4$ , and Tobramycin sulfate were  $\text{C}_4\text{H}_5\text{O}_2^-$  (85  $m/z$ ),  $\text{SO}_3^-$  (80  $m/z$ ), and  $\text{HSO}_4^-$  (97  $m/z$ ). Microprobe imaging uses a finely focused probe beam that is rastered over the desired analysis area. Hence,



**FIGURE 8.16** TOF-SIMS negative ion spectrum of a mixture of the antibiotic Tobramycin sulfate and Simplex-P bone cement, which itself is a mixture of poly(methyl methacrylate) (PMMA) and  $\text{BaSO}_4$ . (Source: Evans Analytical Group, Sunnyvale, CA. Reprinted with permission.)

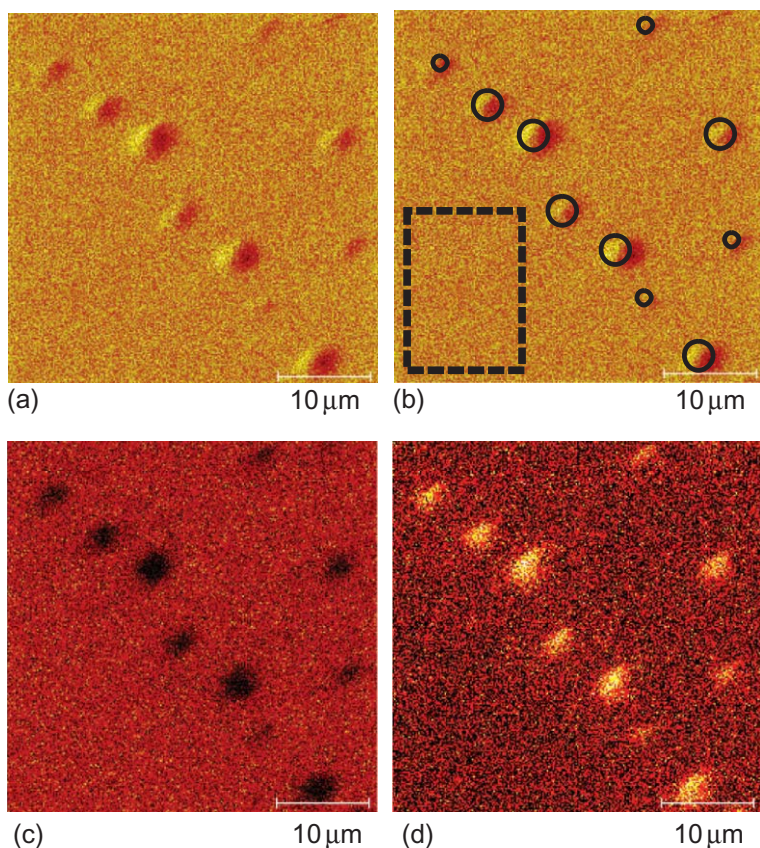


**FIGURE 8.17** Negative ion images showing (a) the general uniformity of the cross-sectioned bone cement sample, and the distributions of (b)  $\text{SO}_3^-$  characteristic of  $\text{BaSO}_4$  and (c)  $\text{HSO}_4^-$  characteristic of Tobramycin sulfate. (Source: Evans Analytical Group, Sunnyvale, CA. Reprinted with permission.)

the primary beam size dictates image resolution. State-of-the-art resolution in microprobe mode is 50 nm diameter.

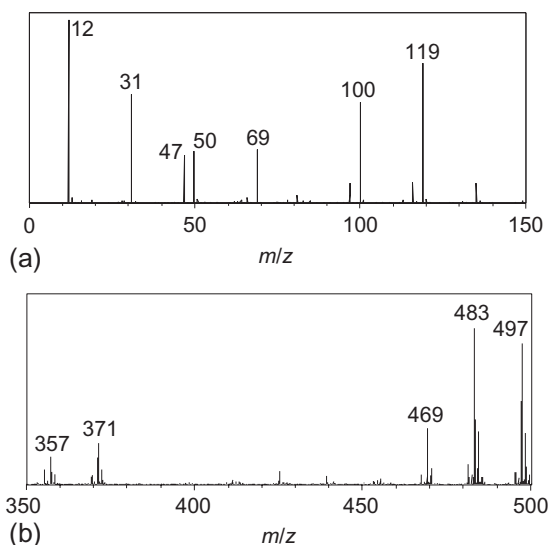
Spectra of organic surfaces can contain literally thousands of peaks. Yet the time required to reach the static limit can be a few minutes or less in small areas. This means that the analyst may not have sufficient time to acquire preliminary spectra necessary for defining ion images without exceeding the static limit. This is particularly true if the sample is both small and unique. In these instances the ability to store all of the raw data is vitally important. Remember that each primary ion pulse generates a complete mass spectrum. Thus, a raw file is a sequence of  $10^4$ – $10^5$  individual spectra. Also included in the raw file are

specifications for the raster pattern from which the  $X$ - $Y$  location of each spectrum can be determined. From this file the analyst can create any desired ion image or display complete mass spectra from any pixel area. The value of this becomes apparent when one inspects spectra of heterogeneous surfaces. Consider a failed hard disk that contained a series of micrometer-sized defects that were visible in optical images of the disk. TOF-SIMS raw data were acquired from an area encompassing the defects. The defects were easily located in the total ion image [see [Figure 8.18\(a\)](#)]. By filtering the data, positive ion spectra from the defects [circled areas in [Figure 8.18\(b\)](#)] and an adjacent defect-free region [boxed area in [Figure 8.18\(b\)](#)] were generated (see [Figure 8.19](#)). Images of intense ions found in the defect-free and defect regions clearly highlight the



**FIGURE 8.18** TOF-SIMS positive ion images of defects on a  $(40\text{-}\mu\text{m})^2$  hard disk surface: (a) total ion image, (b) circled and boxed areas indicating region-of-interest boundaries for defect and control spectra, respectively, (c) image of intense ions (12, 31, 47, and  $50\text{ }m/z$ ) in the defect-free region, and (d) image of intense ions (27, 41, 55, 113, 127, 371, 483, and  $497\text{ }m/z$ ) in the defect regions. (Source: Physical Electronics Inc., Chanhassen, MN. Reprinted with permission.)





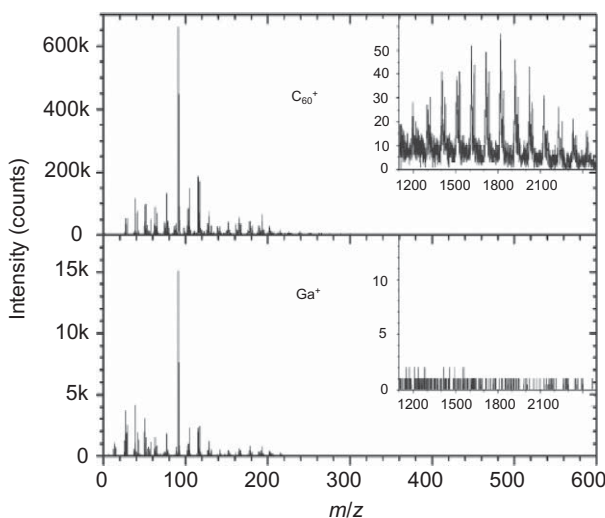
**FIGURE 8.19** Positive TOF-SIMS spectra of (a) the defect-free region and (b) the defect regions on a hard-disk surface. (Source: Physical Electronics Inc., Chanhassen, MN. Reprinted with permission.)

locations of the hydrocarbon droplets in Figure 8.18(c) and (d), respectively. The spectrum from the defect-free area was in excellent agreement with perfluoroether lubricant AM2001, whereas the defect spectrum matched that of a grease used in the drive.

The ultimate quest in TOF-SIMS is to devise a method for quantitative chemical analysis. In order to perform quantitative analysis one must know or be able to predict a material's ion yield, that is, the number of ions formed per incident ion. This is a daunting task since ion yields can vary by several orders of magnitude. Elements with high ionization potentials, such as sodium, tend to more readily form positive ions, whereas those with high electron affinities, such as fluorine, tend to form negative ions. The same theory holds for organic compounds but one must also consider the degree of charge stabilization. For these reasons an aliphatic compound, such as polyethylene, has a positive ion yield many times lower than that of an aromatic ester such as PET. In addition, the matrix from which a compound is ionized can affect ion yields and fragmentation patterns of organic molecules. For example, the presence of sodium ions or silver ions or even organic acids can increase positive ion yields while decreasing fragmentation of organic molecules. For these reasons, many TOF-SIMS studies are purely qualitative. When used with caution, relative ion intensity, such as contaminant ion versus substrate ion, can provide semiquantitative analysis, that is, rank ordering. Truly quantitative analysis is possible if the contaminants and substrate are well defined and control samples of known contaminant concentration can be made. Metal contamination on silicon wafers is one such system.

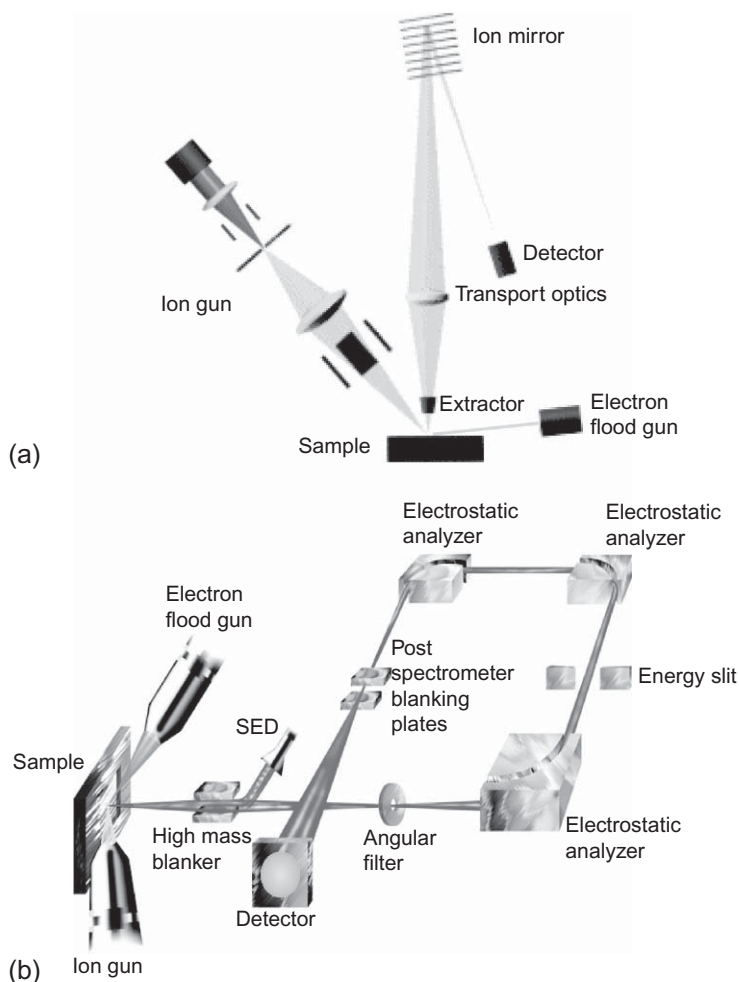
### 8.4.3 TOF-SIMS Instrumentation

A typical TOF-SIMS spectrometer consists of a vacuum chamber operated at a pressure lower than  $10^{-9}$  torr, a pulsed ion gun, extraction and focusing optics, a flight tube, and a detector. Many different ion guns are available such as  $\text{Ga}^+$ ,  $\text{In}^+$ ,  $\text{Ar}^+$ ,  $\text{O}_2^+$ ,  $\text{Cs}^+$ ,  $\text{SF}_5^+$ ,  $\text{Au}_x^+$ ,  $\text{Bi}_x^{2+}$ , and  $\text{C}_{60}^+$ . The two liquid-metal ion guns,  $\text{Ga}^+$  and  $\text{In}^+$ , are easily operated to give high mass resolution (short pulse length) for spectral analysis or high spatial resolution (narrow beam diameter) for microprobe imaging. Gas guns such as  $\text{Ar}^+$  and  $\text{O}_2^+$  are robust and easy to use for large area and/or direct imaging analysis.  $\text{Cs}^+$ , a surface ionization source, is also used for large-area and/or direct imaging analysis and as an auxiliary gun for high depth-resolution profiling. Polyatomic ion guns such as  $\text{SF}_5^+$ ,  $\text{Au}_x^+$ ,  $\text{Bi}_x^{2+}$ , and  $\text{C}_{60}^+$  are used to enhance ion yields of organic compounds while minimizing fragmentation,<sup>64,65</sup> as is clearly demonstrated by comparing the spectra of polystyrene obtained with  $\text{Ga}^+$  and  $\text{C}_{60}^+$  primary ion beams<sup>65</sup> (see Figure 8.20). Flight tubes are of two distinct designs, reflectron<sup>66</sup> and TRIFT<sup>TM</sup>,<sup>67</sup> as shown in Figure 8.21, and are generally up to 2 m long. Other typical components are a computer for instrument control and data acquisition and storage, an electron neutralizer for charge compensation of insulating samples, and a multi-axis sample stage. Cooling and heating stages are useful for preservation of volatile components and sample modification, respectively.



**FIGURE 8.20** Positive TOF-SIMS spectra of poly(styrene) using 10-keV  $\text{C}_{60}^+$  or  $\text{Ga}^+$  ions. Dose (ions):  $\text{C}_{60}^+$ ,  $4.05 \times 10^8$ ;  $\text{Ga}^+$ ,  $4.3 \times 10^8$ . The inset spectra clearly demonstrate that  $\text{C}_{60}^+$  ions produce much higher ion yields for high-mass ions. (Source: Ref. 65. Reprinted with permission.)





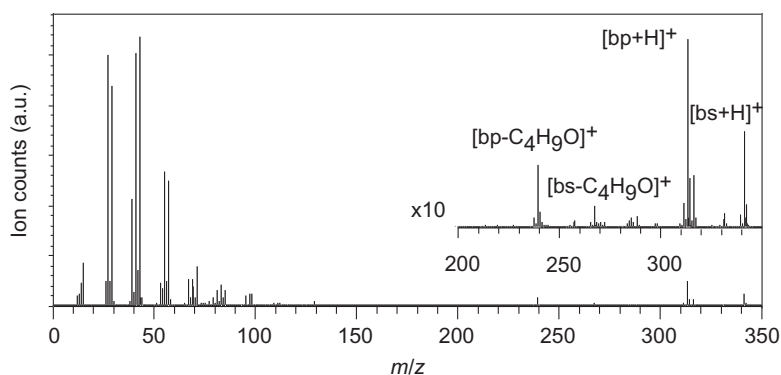
**FIGURE 8.21** Schematic diagrams of (a) a reflection TOF-SIMS system and (b) a TRIFT™ TOF-SIMS system. (Source: Panel (a) ION-TOF GmbH, Münster, Germany and panel (b) Physical Electronics Inc., Chanhassen, MN. Reprinted with permission.)

#### 8.4.4 Applications of TOF-SIMS for Characterizing Surface Contaminants

Identification of contaminants is easiest if the analytical technique can quantitatively detect all elements in the periodic table and all functional groups of organic compounds, or better yet entire molecules, while confining the analytical volume solely to the contaminant. In general, TOF-SIMS fulfills these requirements; however, it is often limited to qualitative and semiquantitative analysis. The shallow sampling depth of the method, one to three atomic layers,

means that the sampling volume is well defined. When combined with the ability of ion images to display graphically the location of contaminants with a lateral resolution down to 50 nm, it becomes quite clear that TOF-SIMS is an excellent contaminant identification tool. It is important to note that any material—organic, inorganic, or metallic—can be analyzed as long as it is vacuum compatible. This generally restricts analysis to solid or low-volatility liquid contaminants on solid substrates.

In many industrial products one wishes to monitor surface contaminants. For example, inspection of raw materials often reveals that bulk contaminant levels can be very low while surface contaminant level can be high. If the material has a low surface area, bulk analytical methods may fail to detect surface contaminants. Moreover, many contaminants and low-molecular-weight molecules preferentially migrate to the outer surface. For example, analysis of starch grains revealed lipids (fats) and other contaminants on the surface.<sup>68</sup> Low surface free energy molecules such as silicones, fatty acids, lipids and their derivatives often migrate to the outer surface of solids, since by doing so the surface free energy of the solid is decreased. Low-molecular-weight molecules such as these are readily identified by TOF-SIMS since the entire molecule is usually removed from the surface intact. This is illustrated in Figure 8.22, where the fatty acid esters butyl palmitate and butyl stearate were found on the outer surface of a molded polypropylene container. In other industrial products one might wish to know the efficacy of different cleaning methods or the effect of processing on surface cleanliness. This is particularly true for orthopedic and dental implants where contamination can lead to infection or premature failure of the implant. Studies with these materials are often performed with model contaminants such as bovine serum albumin or benign bacterial spores to mimic blood or pathogenic contamination, respectively.<sup>69</sup> In these studies molecular fragments are used as markers since the contaminants are too large to be liberated intact.



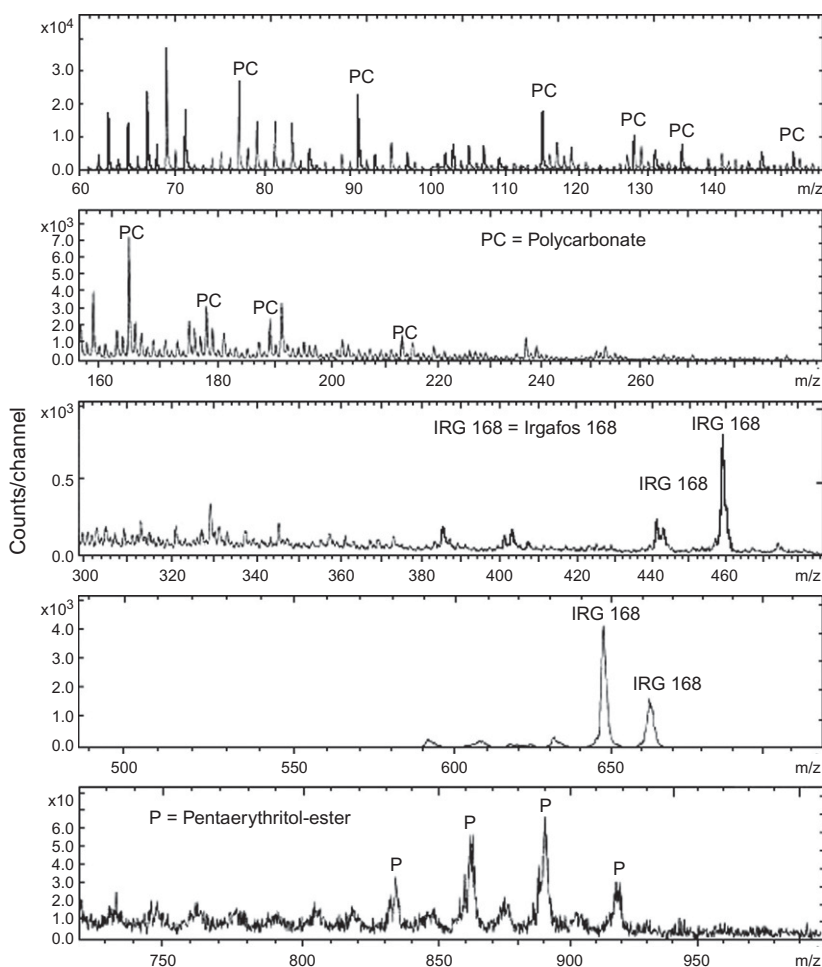
**FIGURE 8.22** Positive TOF-SIMS spectrum of a molded polypropylene container showing contamination by fatty acid esters [butyl palmitate +  $H$ ] $^+$  at 313  $m/z$  and [butyl stearate +  $H$ ] $^+$  at 341  $m/z$ . (Source: Evans Analytical Group, Sunnyvale, CA. Reprinted with permission.)

TOF-SIMS is not limited to analysis of contaminated surfaces. It can also be used to study “pure” contaminants such as those found in dust and air pollution. In a study of 2.5- $\mu\text{m}$  particles from Salt Lake City air pollution,<sup>70</sup> the high spatial resolution of TOF-SIMS permitted analysis of individual particles. The high mass resolution and detection sensitivity of the method enabled trace metal analysis as well as organic compound identification.

Surface contaminants can lead to destruction of the surface, for example, corrosion, or to changes in the properties of the sample, for example, haze formation.<sup>71</sup> In both cases it is important to be able to detect contaminants of all chemical compositions and of varying size. Contaminants also change the chemical properties of surfaces, thereby impacting adhesion and wettability. Poor adhesion and wettability often result in adhesion failure,<sup>72</sup> coating defects,<sup>73</sup> and poor print quality.<sup>74</sup> Contaminants can be introduced to the surface from the environment or they can originate within the material and migrate to the surface.<sup>73</sup> For example, in one study of paint defects in polycarbonate automotive parts, Irgafos 168 and fatty acid esters of pentaerythritol were identified in the defects (see Figure 8.23). While Irgafos 168, an antioxidant added to the polycarbonate, apparently surface segregated, the source of the fatty acid esters was uncertain since it is commonly used as a plasticizer (i.e., an internal source) as well as a release agent (i.e., an external source). TOF-SIMS is widely used to study adhesion failures since many of the contaminants causing adhesion loss have low molecular weight and are readily ionized. The most ubiquitous contaminant that causes adhesion failure is PDMS, which is found in many materials such as anti-foaming agents, cosmetics, hydraulic fluids, lubricants, and release agents. This material has a very high ion yield, and thus the detection limit is generally well below 1% of a monolayer.

High mass resolution and mass range, and small analysis areas, are useful for all types of surface analyses. However, TOF-SIMS affords a unique capability for failure analysis, that is, retrospective analysis. Many failures are one of a kind or at least rare, such that the analyst is often forced to study only a single sample. The ability to perform retrospective data analysis, that is, offline processing of raw data streams, greatly simplifies this task. The analyst need not worry about confining the analysis to the failure area but rather must simply verify that the analysis area includes the failure area and that the appropriate trade-off between mass resolution and spatial resolution has been made. After the raw data are acquired, the analyst can extract from the data ion images to highlight contaminants and mass spectra to ascertain their chemical composition, such as are shown in Figures 8.18 and 8.19. This is particularly important when the analysis area is small, as is often the case in the hard-disk<sup>8</sup> and semiconductor<sup>75</sup> industries.

As stated earlier, TOF-SIMS is often limited to qualitative or semiquantitative analysis. The primary cause of this limitation is the difficulty in producing standards of known contaminant concentrations on well-characterized substrates. One would also like to have alternative analytical methods to corroborate the



**FIGURE 8.23** Positive TOF-SIMS spectrum of a painted polycarbonate automotive part taken inside a coating defect showing contamination by Irgafos 168 and pentaerythritol fatty acid esters. (Source: Ref. 73. Reprinted with permission.)

TOF-SIMS results. One system that meets these requirements is the quantitative surface metal analysis of silicon wafers. Surface metals are routinely studied in situ by total reflection X-ray fluorescence (TXRF), or in solution by vapor phase decomposition inductively coupled plasma mass spectroscopy (VPD-ICPMS) or vapor phase decomposition atomic absorption spectroscopy (VPD-AAS).<sup>29</sup> Were it not for the several potential advantages of TOF-SIMS the methods of choice would remain TXRF, VPD-ICPMS, and VPD-AAS. Whereas TXRF cannot detect hydrogen through fluorine, TOF-SIMS can examine all elements in the periodic table, with equal or better sensitivity.<sup>76–78</sup> VPD-ICPMS and VPD-AAS can also examine all elements; however, unlike TOF-SIMS they are difficult

methods to perform accurately, they cannot differentiate between particulate and plated contaminants, and they do not provide lateral distributions.<sup>77,78</sup> Moreover, TOF-SIMS offers analysis of both inorganic and organic contaminants.<sup>79–81</sup>

Several methods have been developed to generate surface metal calibration standards. The simplest method is to spin coat a solution containing the desired contaminants onto a clean silicon wafer.<sup>82</sup> Unfortunately it is difficult to control the level of contamination and to produce a uniform coverage of metals. A more elegant method is to ion implant metals through a removable layer, with the energy of the implanted species selected to place the peak of the implanted species at the interface between the removable layer and the substrate.<sup>83,84</sup> Once the sacrificial layer is removed, the sample surface contains a known concentration of the implanted species. Contaminant concentration can be easily varied by varying the implant dose. Advantages of this method are the high precision and accuracy of the implant concentration and uniformity across the wafer. The disadvantage is the time and expense in preparing the standards and the need for many different standards. A third method, sputter deposition, has been demonstrated for GaAs wafers.<sup>85</sup> In this method single-element or multi-element targets are sputtered to produce sub-monolayer concentrations of the target elements. As robust protocols for quantification of surface metals contamination by TOF-SIMS are developed, this technique will gain favor as the method of choice.

As one would expect, TOF-SIMS also has been used to verify the efficacy of new cleaning methods. For example, it has been used to study the effect of ozone-based cleaning processes for the removal of organic contaminants from silicon wafers.<sup>86</sup> One prerequisite for studying cleaning efficiency is depositing organic contaminants in a reproducible and quantitative way. Although several methods have been investigated, wet chemical exposure was found to produce the best results.<sup>86</sup> Other studies have used the high sensitivity of TOF-SIMS for organic compounds to verify that cleaning solution additives such as chelating agents did not deposit on wafers during cleaning.<sup>87</sup>

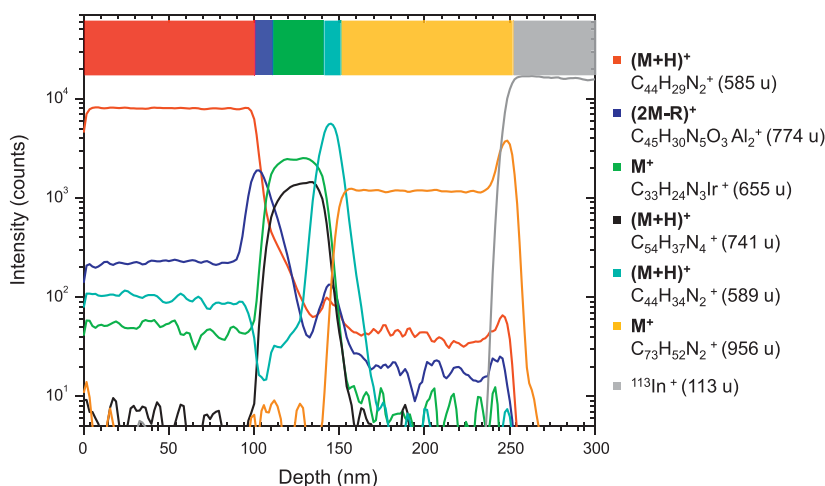
#### 8.4.5 Recent Developments and Future Directions of TOF-SIMS

Advances in the 1990s and early 2000s were driven by the semiconductor industry's need to analyze low concentrations of surface contaminants, primarily surface metals. Tools are now available for analyzing whole 200- and 300-mm wafers. Automation of sample positioning is possible from optical images of the sample or from offline optical detection data (ODD) maps. Advances in both extremes of analysis area now permit the analyst to examine regions as large as 1 mm<sup>2</sup> and as small as 50 nm in diameter.

In the last 10 years, the most significant development of TOF-SIMS instrumentation has been the development of cluster ion sources. Prior to this advancement, TOF-SIMS analysis was limited mostly to surface analysis and depth analysis was only possible on elemental species. It is now possible to

perform molecular species profiling of organic materials. The initial progress in profiling was made possible by polyatomic sources such as  $\text{SF}_5$  and  $\text{C}_{60}$ .<sup>88–90</sup> The low damage cross-section and damage accumulation of these beams enabled organic profiling for tracking of specific molecular species as a function of depth. Although these cluster beams showed promise, there was significant damage observed on certain multistack materials like OLEDs. Currently, the most successful and widely used source for organic profiling is Ar gas cluster ion beam (GCIB). Ar cluster beams are generated with an average Ar cluster size of 1500 atoms.

Early work was driven by molecular simulation studies showing potential benefits using Ar gas cluster ion beams.<sup>91</sup> Comparative studies using  $\text{C}_{60}$  and Ar clusters have shown Ar gas cluster ion beams performed better in terms of low beam-induced chemical damage of surfaces and less roughening of surfaces.<sup>92–97</sup> A study by Moritani et al. showed that Ar gas cluster ion beam has shallow volume sputtering.<sup>95</sup> Figure 8.24 shows organic profiling of a multistack OLED film using an Ar gas cluster ion beam. An  $\text{Ar}_{1700}$  cluster beam was used as the sputtering beam and  $\text{Bi}_3^+$  was used as the analysis beam. One of the main advantages of the Ar gas cluster ion beam is the ability to profile complex multilayer films. An issue commonly encountered in surface analysis using TOF-SIMS is contaminants on the surface, typically due to handling.



**FIGURE 8.24** Depth profile of selected secondary ions using dual beam depth profiling of a multistack OLED film. Analysis beam:  $\text{Bi}_3^+$ . Sputtering beam: 5 keV  $\text{Ar}_{1700}^+$ . NBphen: 2,9-bis(naphthalen-2-yl)-4,7-diphenyl-1,10-phenanthroline; Alq3: Tris-(8-hydroxyquinoline)aluminum; TCTA: Tris(4-carbazoyl-9-ylphenyl)amine; Ir(ppy)3: Tris[2-phenylpyridinato-C<sub>2</sub>,N]iridium(III); Tris[2-phenylpyridinato-C<sub>2</sub>,N]iridium(III); A-NPD: (4,4-bis[N-1-naphthyl-N-phenylamino]biphenyl,  $\alpha$ -NPD); BPAPF: 9,9-bis[4-(N,N-bis(biphenyl-4-yl)amino)phenyl]-9H-fluorene; ITO: indium tin oxide. (Source: Figure courtesy of ION-TOF GmbH.)

The Ar gas cluster ion beam gives the ability to pre-clean the surface and still preserve the substrate chemical information intact.

Fujimoto et al.,<sup>98</sup> in a recently published study, used a combination of Ar gas cluster ion beam TOF-SIMS and liquid chromatography mass spectrometry (LC-MS) techniques to quantify levels of dopant molecules in a multistack OLED structure. This study showed that levels of dopants measured using TOF-SIMS and LC-MS had a linear relationship up to a certain concentration. This study opens the possibility of quantifying species in an organic matrix using TOF-SIMS.

With time, the early limitation of the lack of organic compound reference spectra is slowly being addressed. Current commercial libraries contain several hundred compounds. Still, this is in stark contrast with the profuse reference spectra for other mass spectroscopic methods. Hopefully TOF-SIMS analysts will continue to share reference spectra for the benefit of the entire community. Organic analysis will benefit from larger libraries, as well as from better spectral analysis software. Although expert systems for compound identification are many years away, incremental steps should be taken to ease the burden of examining the thousands of peaks in a typical spectrum. And finally there is the ultimate quest to devise a method for routine quantitative analysis by TOF-SIMS. This quest will require the concerted efforts of analysts, theoreticians, and software engineers, with considerable financial support.

## 8.5 LOW-ENERGY ION SCATTERING

### 8.5.1 Background of LEIS

LEIS, also known as ion scattering spectroscopy (ISS), was first developed in the early 1970s by Smith for use for elemental identification.<sup>99</sup> Since then, it has been used to study the composition and structural properties of the first atomic layer of a sample surface.<sup>100</sup> This technique, based on elastic scattering of atomic ions incident on the surfaces, is one of the four major surface analysis methods including SIMS, AES, and XPS. In contrast to AES and XPS, and somewhat similar to static-SIMS, LEIS only observes the first atomic layer. Its major advantages over other techniques are its extreme surface sensitivity and minimal matrix effects. Because of its rather low spectral resolution between adjacent high-mass elements and poor sensitivity to low-mass elements, LEIS is less used compared to the other three techniques. However, the technique is quite useful for quantitative analysis of the outermost surface layer.

### 8.5.2 Basic Principles of LEIS

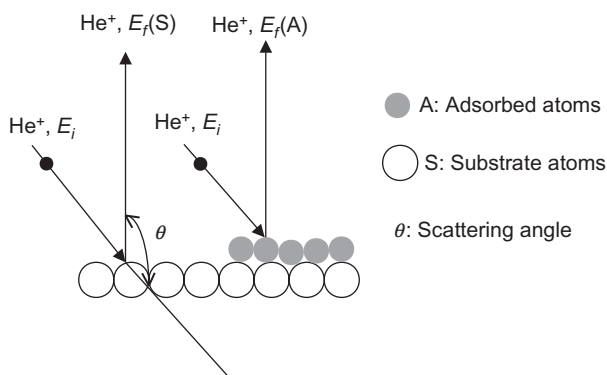
LEIS is a surface analytical technique based on the measurement of backscattered ions during ion-surface collisions.<sup>101</sup> The most commonly used ion is  $^4\text{He}^+$ . When low-energy ions bombard a surface, some of the ions are elastically

scattered from the surface and, consequently, impart some of their kinetic energy to surface atoms. This process is described by a simple two-body elastic collision between an incident ion and target atom and can be modeled as simple hard-sphere or billiard-ball collisions.<sup>102–104</sup> If the incident ions are monoenergetic, having initial kinetic energy  $E_i$ , the kinetic energy  $E_f$  of the scattered ions can be derived from the energy and momentum conservation laws and expressed as

$$E_f = \left[ \frac{\cos \theta + \sqrt{\left(\frac{m_2}{m_1}\right)^2 - \sin^2 \theta}}{1 + \frac{m_2}{m_1}} \right]^2 E_i \quad (8.11)$$

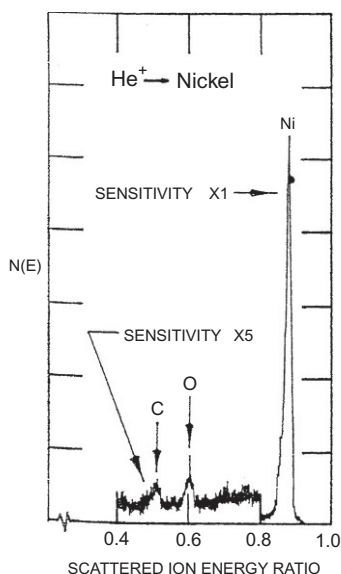
Here  $m_1$  and  $m_2$  are the masses of the incident ions and surface atoms, respectively, and  $\theta$  is the known scattering angle ( $\theta \geq 90^\circ$ ). A schematic of the LEIS process is shown in Figure 8.25. Mass  $m_2$  of the target atom must be greater than  $m_1$  of the incident ion in order to have backscattering. There are no backscattered ions from hydrogen atoms because the mass of the hydrogen target atom will never exceed that of the probe ions. In Equation (8.11),  $E_i$ ,  $m_1$ , and  $\theta$  are fixed. By measuring the kinetic energy  $E_f$  of the scattered ions, one can calculate mass  $m_2$  of the surface target atom. Thus, the kinetic energies of the scattered ions are characteristic of surface atoms.

A typical LEIS spectrum is a plot of the number of scattered ions versus their kinetic energies. A typical LEIS spectrum from a polycrystalline nickel surface<sup>99</sup> is shown in Figure 8.26. This surface was contaminated by an adsorbed hydrocarbon or CO, possibly in combination with a surface oxide. When  $^4\text{He}^+$  probe ions ( $m_1 = 4$ ) bombarded this area, at scattering angle  $\theta$  of  $90^\circ$ , the scattered ion peaks appeared at kinetic energies of  $0.87E_i$ ,  $0.6E_i$ , and  $0.5E_i$ . Thus, according to Equation (8.11), these three peaks corresponded to the scattering from nickel,



**FIGURE 8.25** Schematic of low-energy ion scattering using  $\text{He}^+$  probe ions. (Source: Evans Analytical Group, Sunnyvale, CA. Reprinted with permission.)

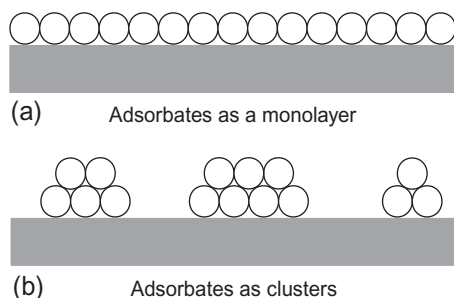




**FIGURE 8.26** Plot of scattered ion intensity versus scattered ion energy ratio ( $E_i/E_o$ ) for 1500 eV  $^4\text{He}^+$  from polycrystalline Ni contaminated with C and O. (Source: Ref. 99. Reprinted with permission.)

oxygen, and carbon surface atoms, respectively. Moreover, this result demonstrates that the ion will retain most of its energy if it is scattered from a heavier atom (like nickel in this example). However, an ion scattered from a lighter atom (such as carbon) will lose most of its energy. Therefore, the LEIS energy spectrum is, in a sense, a mass spectrum. The kinetic energies of the peaks identify each elemental species on the surface. Moreover, peak heights or peak areas can be used to determine surface elemental concentrations.<sup>104</sup>

The extreme surface sensitivity of LEIS results from the high neutralization probability of the  $\text{He}^+$  ions.<sup>105–107</sup> As a consequence, only  $\sim 10^{-3}$  to  $10^{-4}$  of the primary ions will survive the collision and finally reach the detector. The scattered ions originating from deeper than the first atomic layer have a high probability of being neutralized and are not detectable. This makes LEIS unique in studying the surface composition of the first atomic layer. A systematic investigation of metal/metal oxide systems using LEIS clearly illustrates its capability of distinguishing a monolayer of adsorbed atoms from those present in clusters.<sup>108–110</sup> If a surface is completely covered by a monolayer of adsorbed atoms, as shown in Figure 8.27(a), only the ions scattered from the adsorbate are measured and the substrate intensity is completely attenuated. However, if the same amount of adsorbate forms clusters [Figure 8.27(b)], a fraction of the substrate will remain exposed, and ions scattered from both the adsorbate and substrate will be detected.



**FIGURE 8.27** Schematic diagram of adsorbed atoms as: (a) a monolayer versus (b) clusters. (Source: Evans Analytical Group, Sunnyvale, CA. Reprinted with permission.)

As stated earlier, an LEIS spectrum is essentially a mass spectrum. Each peak represents an element in the surface layer. Of equal importance is the fact that peak intensities can be used to determine elemental concentrations, therefore LEIS can be used for quantitative analysis. Strictly speaking, since one measures atoms from the first atomic layer, LEIS can detect only areal densities rather than volume concentrations as in AES and XPS. The intensity of the ions scattered from atom  $i$  into angle  $\theta$  can be described as<sup>111</sup>

$$I_i = I_o \sigma_i P_i^+ R T n_i \quad (8.12)$$

where  $I_o$  is incident ion current,  $\sigma_i$  is the differential scattering cross-section for scattering from element  $i$ ,  $P_i^+$  is the ion survival probability,  $R$  is a factor that accounts for surface roughness related to the shadowing and blocking effects,  $T$  is the transmission function of the spectrometer, and  $n_i$  is the number of surface atoms  $i$  per unit area. Thus, the areal density  $n_i$  can, in principle, be determined from the measured intensity  $I_i$  by

$$n_i = \frac{I_i}{I_o \sigma_i P_i^+ R T} \quad (8.13)$$

It should be noted that direct quantification is possible only if all the parameters on the right-hand side of Equation (8.13) are known. However, this method is not generally used because parameters, such as  $P_i^+$  and  $R$ , are poorly known and result in large experimental errors.

In practice, quantitative analysis of surface elemental concentration is based on calibration against standard compounds. Equation (8.12) can be rewritten as

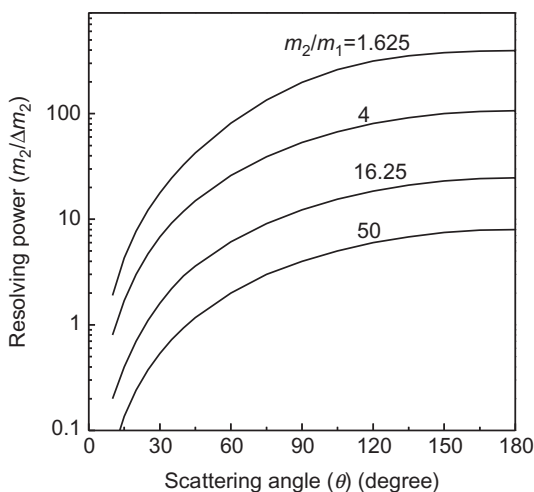
$$I_i = I_o S_i c_i \quad (8.14)$$

where  $S_i$  is the elemental sensitivity factor relative to a standard and  $c_i$  is the relative surface concentration of element  $i$ . Therefore, the relative surface concentration, that is, surface coverage of element  $i$ , can be obtained as

$$c_i = \frac{I_i}{I_o S_i} \quad (8.15)$$

It has been found that  $S_i$  generally does not depend on the chemical environment of an atom in the surface (i.e., there are few or no matrix effects). This makes it relatively simple to perform quantitative analysis in LEIS. Relative sensitivity factors  $S_i$  have been measured from pure compounds by different laboratories through round-robin experiments, which allow straightforward calculation of relative elemental concentrations. Furthermore, it has been noted that  $S_i$  increases monotonically with atomic mass.<sup>100</sup> Because of this simple relationship, it is possible to use the peak intensity as a direct quantitative measurement of the elemental concentration.

Theoretically, LEIS can be used to detect all elements having atomic mass greater than that of the incident ions. Its ability to resolve the masses of different atoms depends on the energy spread of the incident ions, the mass ratio  $m_2/m_1$  between the target atoms and incident ions, and the scattering angle  $\theta$ . The optimum mass resolution can be obtained by choosing the mass ratio  $m_2/m_1$  to be close to unity, as shown in Figure 8.28.<sup>30</sup> Moreover, a large scattering angle also gives good mass resolution. For low-mass atoms, LEIS has good mass resolution but poor sensitivity with minimum detection levels of a few at. %. For higher-mass atoms, this technique has very good sensitivity but poor element specificity. For example, LEIS can be used to detect ppm levels of gold, but it is extremely difficult to resolve adjacent high-mass atoms, such as Au-Pt, or even Cu-Zn, Fe-Co, K-Ca, etc. The elemental specificity for high-mass atoms can be improved by using higher-mass probe ions ( $\text{Ne}^+$ ,  $\text{Ar}^+$ ,  $\text{Kr}^+$ , and  $\text{Xe}^+$ ).



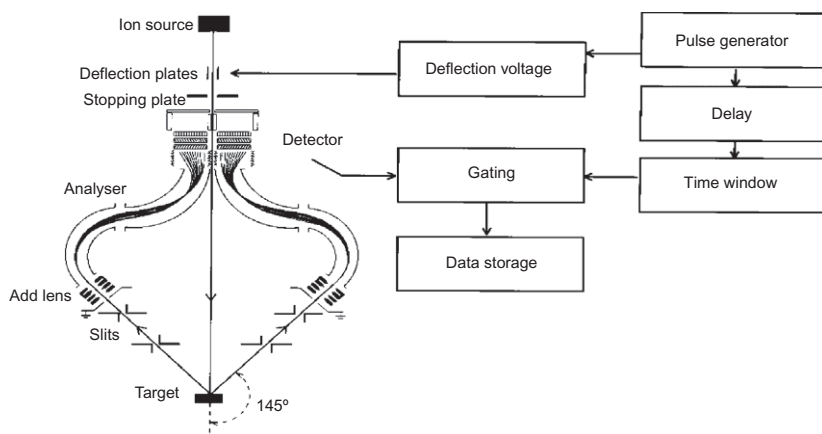
**FIGURE 8.28** Mass resolving power versus the scattering angle for different mass ratios of  $m_2/m_1$  (plot reproduced according to Equation (4.13) in Ref. 30). (Source: Evans Analytical Group, Sunnyvale, CA. Reprinted with permission.)



### 8.5.3 LEIS Instrumentation

The basic elements for LEIS experiments are an ultrahigh vacuum chamber, an ion gun, and an energy analyzer. This technique is often designed as an add-on technique to supplement XPS and AES. A differentially pumped ion gun is used to generate positive ions with low energies, typically from 500 eV to 5 keV. The most commonly used ion sources are the noble gas ions  $\text{He}^+$ ,  $\text{Ne}^+$ , and  $\text{Ar}^+$ , or the alkali metal ions  $\text{Li}^+$ ,  $\text{Na}^+$ , and  $\text{K}^+$ . In most instances the spatial resolution of LEIS is limited by the size of the ion beam. The same energy analyzers as used for XPS and AES, that is, cylindrical mirror or CHAs, can also be used for LEIS. Only the analyzer polarity needs to be changed because the ions are positively charged. Large scattering angles are needed in most LEIS experiments (e.g., they range from  $130^\circ$  to  $170^\circ$ ) in order to obtain maximum resolving power.

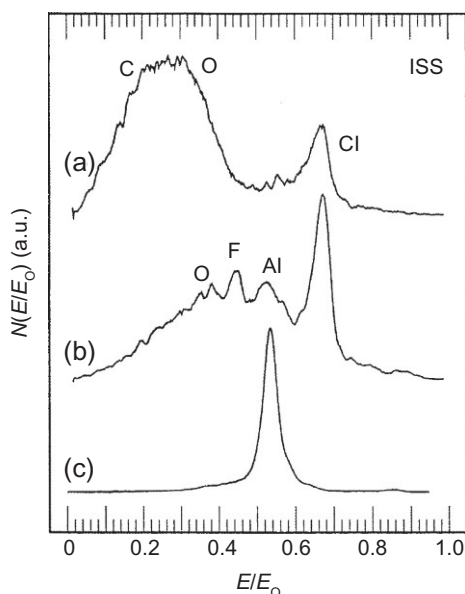
When using conventional LEIS instrumentation, the probe ions often cause damage to the sample surface. Ion-induced damage includes the sputtering of atoms from the surface and breaking of surface chemical bonds. To circumvent this problem, Brongersma and coworkers designed a parallel detection analyzer to maximize data collection while minimizing analysis time.<sup>112</sup> As shown in Figure 8.29, a double-toroidal electrostatic analyzer is used for imaging the energy distribution onto a position-sensitive detector.<sup>112</sup> A specially designed position-sensitive detector allows simultaneous detection of energies from 100 channels and has sensitivity about 1000 times that of conventional LEIS measurements. This increased detection efficiency allows one to use very low ion beam current, therefore beam damage is significantly reduced.



**FIGURE 8.29** Schematic diagram of toroidal electrostatic analyzer with a position-sensitive detector. (Source: Ref. 100. Reprinted with permission.)

### 8.5.4 Applications of LEIS for Characterizing Surface Contaminants

In many applications, surface contaminants are concentrated as the first atomic layer of a surface. The end result is that the elemental concentration of the outermost surface will be significantly different from that in deeper layers. Since LEIS is unique in studying elemental compositions of the first atomic layer due to its extreme surface sensitivity, it is well suited for these studies. A particular strength of the method is the ability to perform quantitative analysis of the outermost surface layer. The early applications of LEIS included the detection of surface contaminants on pure conductive materials, particularly on metal and alloy surfaces. Because these materials have high surface free energies, adventitious carbon contamination is almost always present. Moreover, exposure to air and moisture often causes the formation of metal oxides and hydroxides on the surface. Since LEIS can easily distinguish low-mass elements such as carbon and oxygen from high-mass transition metals, it is well suited for studying this type of samples. As is shown in Figure 8.25, LEIS unambiguously identified carbon and oxygen on a polycrystalline nickel surface contaminated with adsorbed hydrocarbon or CO, possibly in combination with a surface oxide. In addition, LEIS studies of aluminum foils<sup>113</sup> further demonstrate its ability to identify surface impurities on metal surfaces. As displayed in Figure 8.30(a),<sup>113</sup> the LEIS



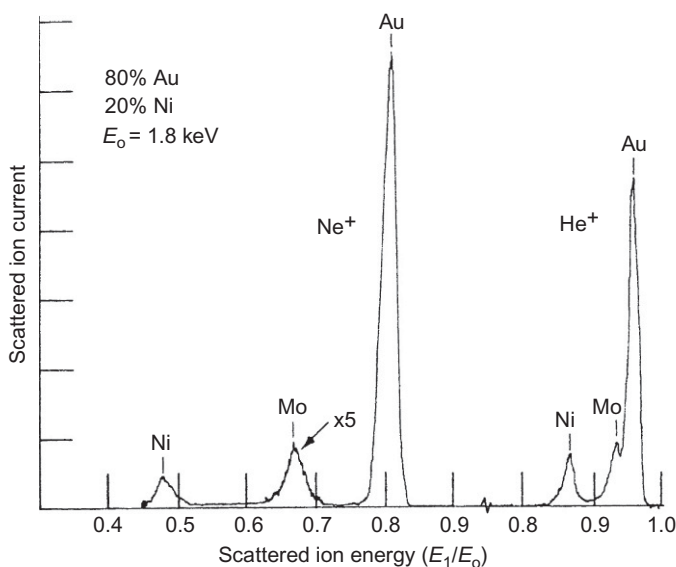
**FIGURE 8.30** LEIS spectra taken from the hot-rolled Al foil using 1 keV  $\text{He}^+$  after (a) insertion into the UHV chamber, (b) a 12-h exposure to the H-atom flux, and (c) 4-h sputtering with 2 keV  $\text{Ne}^+$ . (Source: Ref. 113. Reprinted with permission.)

spectrum taken from an as-received hot-rolled aluminum foil indicates that the outermost surface was contaminated by carbon, oxygen, and chlorine. Aluminum was not detected in this spectrum, which implied that aluminum lay beneath the organic layer.

LEIS is also used to evaluate the efficiency of surface cleaning methods. Since surface contaminants often interfere with product operation and even cause product failures, they need to be removed during processing. In the study of hot-rolled aluminum foil it was found that a 12-h exposure of the surface to a flux of hydrogen atoms removed the carbon and oxygen contamination, as shown in Figure 8.30, spectrum b. Subsequent  $\text{Ne}^+$  sputtering for 4 h removed almost all the remaining surface contaminants and left a clean aluminum surface, as shown in Figure 8.30, spectrum c. Similarly, an air-exposed polycrystalline Ni/Cr alloy was studied by LEIS.<sup>114</sup> The outermost surface of the as-received Ni/Cr alloy not only contained nickel, but also had oxygen, carbon, and sodium impurities. Following a 12-h exposure to a hydrogen atom flux and a subsequent 15-min  $\text{Ar}^+$  sputtering, a clean alloy surface with only a small amount of oxygen was obtained.<sup>114</sup> In practice, for these near-surface contaminants, not only can LEIS provide quantitative evaluation of the surface cleanliness, but it can verify that new contaminants were not introduced during the cleaning process. For instance, thermal heating is a gentle method to remove organic compounds from inorganic surfaces, but can cause substrate modification and damage. An example was given by Smith, who studied Au-Ni alloys.<sup>115</sup> After heat treatment at 250 °C, molybdenum impurities were found to segregate to the surface. In this particular case the use of an  $\text{Ne}^+$  probe rather than the more conventional  $\text{He}^+$  probe is beneficial since it permits complete separation of molybdenum and gold (see Figure 8.31).<sup>115</sup>

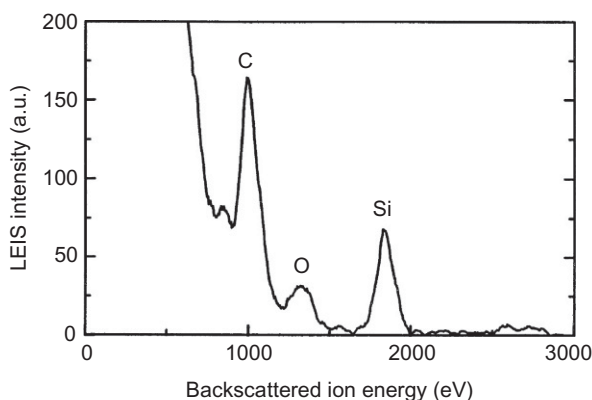
The study of corrosion on finished metals is of great interest. Corrosion not only results in cosmetic blemishes and surface damage but can cause adhesion failure. Sometimes, even fractional monolayer contamination by elements such as chlorine and sulfur can initiate corrosion. The application of LEIS to monitor outermost surface contaminants has been extremely helpful in providing critical information for device manufacture.<sup>116</sup> Examples have included the detection of surface contaminants on copper,<sup>116</sup> steel,<sup>117</sup> and electrical contacts.<sup>100</sup> Depth profiling by ion beam sputtering was also utilized during these experiments to measure the in-depth composition and thickness of the contaminants. As in SIMS, AES, and XPS, the etching capability of ion beams allows depth profiling measurement in LEIS. Either a single-ion beam is used for both etching and LEIS analysis, or dual-ion beams are used with one for etching and the other for LEIS analysis. A depth profile plot in LEIS represents areal density at each sputter depth versus sputter time. If the sputter rate is known, a depth profile can be converted to elemental concentration versus sputter depth.<sup>116,117</sup>

The study of surface contaminants and segregation on polymer surfaces is another area that has generated considerable interest recently. As indicated previously, the LEIS system designed by Brongersma and coworkers makes it



**FIGURE 8.31** LEIS spectra using  $\text{He}^+$  and  $\text{Ne}^+$  ions from Ni-Au alloy contaminated by Mo. Note the increased mass resolution obtained by choosing the heavier primary ions ( $\text{Ne}^+$ ). (Source: Ref. 115. Reprinted with permission.)

possible to study polymers without causing noticeable beam damage to the surface.<sup>112</sup> LEIS is particularly useful for studying surface segregation, when often a monolayer or less of surface contaminants are present. A good demonstration of its unique capability can be seen by studying poly(3-hexylthiophene) (P3HT) films.<sup>118</sup> In this example, LEIS was used to investigate P3HT films that had 2% siloxane monomers in the bulk. The topmost surface was found to be silicone enriched due to the segregation of siloxane towards the outermost surface. It contained 25-100% siloxane (depending on the preparation method) by referencing to a pure PDMS. In addition, sputter profiling was also performed on these samples to determine the siloxane chain orientation. LEIS depth profile results indicated that the methyl groups of the siloxane were preferentially situated on the outermost surface of P3HT and partially screened oxygen atoms that were below the methyl groups. Furthermore, by comparing the measured and calculated LEIS intensities of surface carbon, oxygen, and sulfur at different surface siloxane coverage, it was concluded that the structure of the siloxane on P3HT resembled that of a PDMS film. Another study examined the outermost surface of a poly(phenylene vinylene) (PPV) semiconducting polymer in a polymer light-emitting diode (PLED).<sup>110</sup> A strong silicon peak was observed in the LEIS spectrum and indicated that the surface was heavily contaminated by silicone, as displayed in Figure 8.32.<sup>110</sup> This silicone contamination was caused by the surface segregation of silicone from the bulk PPV layer.



**FIGURE 8.32** LEIS spectrum from a poly(phenylene vinylene) (PPV) semiconducting polymer using 3 keV  $^3\text{He}^+$  ions. Note that the strong silicon peak shows that the surface is heavily contaminated by siloxane. (Adapted from Ref. 110. Reprinted with permission.)

### 8.5.5 Recent Developments and Future Directions of LEIS

In comparison with other surface analysis techniques, LEIS has its merits as well as disadvantages. As has been demonstrated, it is conceptually simple, quantitative, and extremely surface sensitive. However, this technique has relatively poor elemental specificity for materials consisting of elements with similar masses. Furthermore, in studying polymer surfaces, the probe ions often cause damage to the surface. This is particularly true for older instruments. Therefore, its applications have been limited compared to other surface analysis techniques. Fortunately, progress has been made since the 1990s, and now it is possible to analyze samples without causing significant damage to the surface. This is particularly important when studying polymer surfaces.

Based on its past applications, the future trends of LEIS will be focused on improvements in sensitivity, mass resolution, and analysis time. As it becomes more widely recognized that the composition of the first atomic layer is critical in a product's performance, LEIS will likely become a more common technique in routine surface analysis. One hopes that commercially available systems that offer higher performance and ease of use will be available in the near future.

## 8.6 SUMMARY

When selecting a method for surface contaminant analysis one must consider factors such as the size, shape, morphology, composition, volatility, concentration, and history of both the substrate and the contaminant. In addition one must decide whether the study should provide qualitative or quantitative results and what specific type of information is required, for example, elemental composition or



chemical bonding or exact compound identification. AES, XPS, TOF-SIMS, and LEIS are four classical surface analysis methods, each capable of quantitative analysis of sub-monolayer contaminants.

These methods differ significantly in applicability for particle analysis. LEIS and XPS require particles of 100 and 10  $\mu\text{m}$  diameters, respectively. In contrast, sub-micrometer particle analysis requires the use of TOF-SIMS or AES, with AES having the smallest analysis size of 10 nm diameter. Whenever elemental analysis is insufficient for contaminant identification, XPS and TOF-SIMS are the methods of choice. This is particularly true for analysis of organic materials. Often the choice between XPS and TOF-SIMS can be made based on whether the contaminant is a complete unknown, for which XPS excels, or whether a specific contaminant needs to be identified, for which TOF-SIMS excels. Minimum detection limit is often a deciding factor in method selection. LEIS and AES generally have limited sensitivity with XPS being only slightly more sensitive. On the other hand, TOF-SIMS has extremely low detection limits for all elements and many organic molecules, particularly those having molecular weights between  $\sim 100$  and  $\sim 500$  amu. In many instances the choice of analysis method is not immediately obvious. It is then helpful to consult [Table 8.1](#) to rank the suitability of these methods. Given the complexity of manufacturing processes and the often uncontrolled history of industrially relevant samples, one is forced frequently to employ multiple analytical methods.

## NOTE

- a. For one case in a round-robin experiment performed on a set of representative polycrystalline, metallic samples at five different laboratories, Cu was used as a standard.<sup>100</sup>

## REFERENCES

1. Auger, P. Sur l'effet photoélectrique composé. *J. Phys. Radium* **1925**, 6, 205.
2. Lander, J. J. Auger Peaks in the Energy Spectra of Secondary Electrons from Various Materials. *Phys. Rev.* **1953**, 91, 1382.
3. Harris, L. A. Analysis of Materials by Electron-Excited Auger Electrons. *J. Appl. Phys.* **1968**, 39, 1419.
4. Harris, L. A. Some Observations of Surface Segregation by Auger Electron Emission. *J. Appl. Phys.* **1968**, 39, 1428.
5. Briggs, D.; Seah, M. P., Eds. *Auger and X-ray Photoelectron Spectroscopy*, vol. 1; *Practical Surface Analysis*; 2nd ed.; John Wiley: Chichester, UK, 1990.
6. Sanders, J. H. Investigation of Grain Boundary Chemistry in Al-Li 2195 Welds Using Auger Electron Spectroscopy. *Thin Solid Films* **1996**, 277, 121.
7. Vezeau, P. J.; Koorbusch, G. F.; Draughn, R. A.; Keller, J. C. Effects of Multiple Sterilization on Surface Characteristics and In Vitro Biologic Responses to Titanium. *J. Oral Maxillofac. Surg.* **1996**, 54, 738.
8. Yang, M. C.; Luo, J. B.; Wen, S. Z.; Wang, W.; Huang, Q.; Hao, C. W. Failure Characterization at Head/Disk Interface of Hard Disk Drive. *Sci. China Ser. A Math. Phys. Astron.* **2001**, 44(Suppl S), 407.

9. Uritsky, Y.; Chen, L.; Zhang, S.; Wilson, S.; Mak, A.; Brundle, C. R. Root Cause Determination of Particle Contamination in the Tungsten Etch Back Process. *J. Vac. Sci. Technol. A* **1997**, *15*, 1319.
10. Smadi, M. M.; Kong, G. Y.; Carlile, R. N.; Beck, S. E. Particle Contamination of Silicon in SF<sub>6</sub> and CF<sub>4</sub>/O<sub>2</sub> RF Etch Plasmas. *J. Electrochem. Soc.* **1992**, *139*, 3356.
11. Ernst, K. H.; Grman, D.; Hauert, R.; Hollander, E. Fluorine-Induced Corrosion of Aluminum Microchip Bond Pads—An XPS and AES Analysis. *Surf. Interface Anal.* **1994**, *21*, 691.
12. Chan, L. H.; Altmanand, E. I.; Liang, Y. Development of Procedures for Obtaining Clean, Low-Defect-Density Ge(100) Surfaces. *J. Vac. Sci. Technol. A* **2001**, *19*, 976.
13. Lee, K. N.; Donovan, S. M.; Gila, B.; Overberg, M.; Mackenzie, J. D.; Abernathy, C. R.; Wilson, R. G. Surface Chemical Treatment for the Cleaning of AlN and GaN Surfaces. *J. Electrochem. Soc.* **2000**, *147*, 3087.
14. Bertru, N.; Bacquet, Y.; Nouaoura, M.; Lassabatere, L. UV-O<sub>3</sub> Preparation of InAs (100) Surfaces Prior to MBE Growth. *Appl. Surf. Sci.* **1994**, *74*, 331.
15. Lu, Y. F.; Song, W. D.; Hong, M. H.; Ren, A. M.; Chen, Q.; Chong, T. C. Laser Cleaning of IC Mould and Its Real-Time Monitoring. *Jpn. J. Appl. Phys.* **2000**, *39 Part 1*, 4811.
16. Feng, Y.; Liu, Z.; Vilar, R.; Yi, X. S. Laser Surface Cleaning of Organic Contaminants. *Appl. Surf. Sci.* **1999**, *150*, 131.
17. Larciprete, R.; Borsella, E. Excimer Laser Cleaning of Si(100) Surfaces at 193 and 248 nm Studied by LEED, AES and XPS Spectroscopies. *J. Electron Spectrosc. Rel. Phenom.* **1995**, *76*, 607.
18. Sung, M. M.; Lee, S. H.; Lee, S. M.; Marton, D.; Perry, S. S.; Rabalais, J. W. Composition and Morphology of InP{100} Surfaces as a Function of Low Energy Ar<sup>+</sup> Bombardment and Annealing. *Surf. Sci.* **1997**, *382*, 147.
19. Chong, Y. F.; Gopalakrishnan, R.; Tsang, C. F.; Sarkar, G.; Lim, S.; Tatti, S. An Investigation on the Plasma Treatment of Integrated Circuit Bond Pads. *Microelectron. Reliab.* **2000**, *40*, 1199.
20. Fillmore, D. K.; Krasinski, H. A. Evaluation of an Auger Electron Spectrometer for 200 mm Semiconductor Defect Review. *Surf. Interface Anal.* **1998**, *26*, 109.
21. Jacka, M. Scanning Auger Microscopy: Recent Progress in Data Analysis and Instrumentation. *J. Electron Spectrosc. Rel. Phenom.* **2001**, *114*, 277.
22. Hertz, H. Ueber einen Einfluss des ultravioletten Lichtes auf die elektrische Entladung. *Ann. Phys.* **1887**, *267*, 983.
23. Einstein, A. Über einen die Erzeugung und Verwandlung des Lichtes betreffenden heuristischen Gesichtspunkt. *Ann. Phys.* **1905**, *322*, 132.
24. Steinhardt, R. G. X-Ray Photoelectron Spectrometer for Chemical Analysis. *Anal. Chem.* **1951**, *23*, 1585.
25. Nordling, C.; Sokolowski, E.; Siegbahn, K. Precision Method for Obtaining Absolute Values of Atomic Binding Energies. *Phys. Rev.* **1957**, *105*, 1676.
26. Siegbahn, K.; Nordling, C.; Sokolowski, E. Chemical Shifts of Photo- and Auger Electron Lines. In: Lipkin R. H. Ed.; *Proc. 1957 Rehovot Conf. Nuclear Structure*; North-Holland: Amsterdam, The Netherlands, 1958; p 291.
27. Sokolowski, E.; Nordling, C.; Siegbahn, K. Chemical Shift Effect in Inner Electronic Levels of Cu Due to Oxidation. *Phys. Rev.* **1958**, *110*, 776.
28. Nordling, C.; Sokolowski, E.; Siegbahn, K. Evidence of Chemical Shifts of the Inner Electronic Levels in a Metal Relative to Its Oxides (Cu, Cu<sub>2</sub>O, CuO). *Arkiv Fysik* **1958**, *13*, 483.
29. Brundle, C. R.; Evans, C. A.; Wilson, S., Eds. *Encyclopedia of Materials Characterization*; Butterworth-Heinemann: Boston, MA, 1992.

30. Woodruff, D. P.; Delchar, T. A. *Modern Techniques of Surface Science*, 2nd ed.; Cambridge University Press: Cambridge, UK, 1994.
31. Barr, T. L. *Modern ESCA*; CRC Press: Boca Raton, FL, 1994.
32. Beamson, G.; Briggs, D. *High Resolution XPS of Organic Polymers*, John Wiley: Chichester, UK, 1992.
33. Cumpson, P. J. Angle-Resolved XPS and AES: Depth Resolution Limits and a General Comparison of Properties of Depth-Profile Reconstruction Methods. *J. Electron Spectrosc. Rel. Phenom.* **1995**, 73, 25.
34. Ertl, G.; Küppers, J. *Low Energy Electrons and Surface Chemistry*, 2nd ed.; Verlag Chemie: Weinheim, Germany, 1985.
35. Ilten, K.; MacDonald, B.; Brox, O.; Benninghoven, A.; Weiss, C.; Hossain, T.; Zschech, E. Ultra-Shallow Junction Depth Profile Analysis Using TOF-SIMS and TXRF. AIP Conference Proceedings 449, In: *Characterization and Metrology for ULSI Technology*; Seiler, D. G., Diebold, A. C., Bullis, W. M., Shaffner, T. J., McDonald, R., Walters, E. J., Eds.; AIP Press: New York, NY, 1998; p 777.
36. Wagner, C.D. *Electron Spectroscopy*. North-Holland: Amsterdam, The Netherlands, 1972; p. 861.
37. Moretti, G. Auger Parameter and Wagner Plot in the Characterization of Chemical States by X-ray Photoelectron Spectroscopy: A Review. *J. Electron Spectrosc. Rel. Phenom.* **1998**, 95, 95.
38. Galuska, A. A.; Halverson, D. E. Quantitative Analysis of Surface Ethylene Concentrations in Ethylene-Propylene Polymers using XPS Valence Bands. *Surf. Interface Anal.* **1998**, 26, 425.
39. Wolan, J. T.; Hoflund, G. B. Surface Characterization Study of AgF and AgF<sub>2</sub> Powders Using XPS and ISS. *Appl. Surf. Sci.* **1998**, 125, 251.
40. Keller, J. C.; Wick, D. G.; Draughn, R. A.; Wightman, J. P. The Effects of Sterilization Treatments on Adhesion of Bone Cells to Titanium. *J. Adhes.* **1995**, 54, 145.
41. Cano, E.; Simancas, J.; Bastidas, J. M. Corrosion of Electrolytically Gilded Silver: A Case Study. *Plat. Surf. Finish.* **2001**, 88(1), 80.
42. Battistoni, C.; Cantelli, V.; Debenedetti, M.; Kaciulis, S.; Mattogno, G.; Napoli, A. XPS Study of Ceramic Three-Way Catalysts. *Appl. Surf. Sci.* **1999**, 144–145, 390.
43. Brewis, D. M.; Critchlow, G. W. Adhesion and Surface Analysis. *J. Adhes.* **1995**, 54, 175.
44. Cherian, J. T.; Castner, D. G. Using ESCA and SIMS in Composites Manufacturing. *J. Adv. Mater.* **2000**, 32, 28.
45. Application Note BN 1423. *XPS Analysis of Disposable Gloves*; Evans Analytical Group, 2006. <http://www.eag.com/documents/disposable-gloves-xps-analysis-BN1423.pdf>.
46. Moffitt, C. E.; Reddy, C. M.; Yu, Q. S.; Wieliczka, D. M.; Yasuda, H. K. Selective Adsorption of Fluorocarbons and Its Effects on the Adhesion of Plasma Polymer Protective Coatings. *Appl. Surf. Sci.* **2000**, 161, 481.
47. Braski, D. N.; Winters, E. D. Auger Electron Spectroscopy and X-Ray Photo-electron Studies of Adhesion Failure of Ni/Au-Plated Contacts in Ceramic Electronic Packages. *J. Vac. Sci. Technol. A* **1996**, 14, 1348.
48. Wadsak, M.; Astrup, T.; Wallinder, I. O.; Leygraf, C.; Schreiner, M. Multianalytical In Situ Investigation of the Initial Atmospheric Corrosion of Bronze. *Corros. Sci.* **2002**, 44, 791.
49. Colin, S.; Beche, E.; Berjoan, R.; Jolibois, H.; Chambaudet, A. An XPS and AES Study of the Free Corrosion of Cu-, Ni- and Zn-Based Alloys in Synthetic Sweat. *Corros. Sci.* **1999**, 41, 1051.
50. Xu, C. H.; Gao, W.; Hyland, M.; Gong, H. Characterization of High Temperature Corrosion Products on FeAl Intermetallics by XPS. *Corros. Sci.* **2001**, 43, 1891.

51. Brownson, R.; Butler, K.; Cadena, S.; Detar, M.; Johnson, I.; McCouulloch, L.; McCouulloch, J.; Mishra, B.; Healey, J.; Honcik, K.; Phan, T.; Sterif, T.; Stevens, H.; Tovar, A. The Role of RIE in Microchip Bond Pad Corrosion. In: *Proceedings of SPIE 2874* 1997; pp 95–102.
52. Ochs, D.; Dieckhoff, S.; Cord, B. Characterization of Hard Disk Substrates (NiP/Al, Glass) Using XPS. *Surf. Interface Anal.* **2000**, 30, 12.
53. Kamal, T.; Hess, D. W. Photoresist Removal Using Low Molecular Weight Alcohols. *J. Electrochem. Soc.* **2000**, 147, 2749.
54. O’Kane, D. F.; Mittal, K. L. Plasma Cleaning of Metal Surfaces. *J. Vac. Sci. Technol.* **1974**, 11, 567.
55. Fazlin, F. Plasma Treatment for Improved Wire Bonding. *Solid State Technol.* **1996**, 39(10), 117.
56. So, S. K.; Chio, W. K.; Cheng, C. H.; Leung, L. M.; Kwong, C. F. Surface Preparation and Characterization of Indium Tin Oxide Substrates for Organic Electroluminescent Devices. *Appl. Phys. A* **1999**, 68, 447.
57. Deng, Z. B.; Ding, X. M.; Liao, L. S.; Hou, X. Y.; Lee, S. T. The Interface Analyses of Inorganic Layer for Organic Electroluminescent Devices. *Displays* **2000**, 21, 79.
58. Application Notes. *The PHI Quantum 2000: A Scanning ESCA Microprobe*; Physical Electronics: Eden Prairie, MN, 1997.
59. Technology Notes. *Axis Ultra: High Performance Imaging XPS*; Kratos Analytical: Manchester, UK, 1998.
60. Zubritsky, E. XPS Up Close. *Anal. Chem.* **2001**, 73, 297A.
61. Application Notes. *Overview: A Brief Introduction to Parallel X-Ray Photoelectron Imaging on the ESCALAB 250*; Thermo VG Scientific: Waltham, MA, 2001.
62. Benninghoven, A. Analysis of Submonolayers on Silver by Negative Secondary Ion Emission. *Phys. Status Solidi* **1969**, 34, K169.
63. Wilson, R. G.; Stevie, F. A.; Magee, C. W. *Secondary Ion Mass Spectrometry: A Practical Handbook for Depth Profiling and Bulk Impurity Analysis*; John Wiley: New York, NY, 1989.
64. Kotter, F.; Benninghoven, A. Secondary Ion Emission from Polymer Surfaces under Ar<sup>+</sup>, Xe<sup>+</sup>, and SF<sub>5</sub><sup>+</sup> Ion Bombardment. *Appl. Surf. Sci.* **1998**, 133, 47.
65. Weibel, D.; Wong, S.; Lockyer, N.; Blenkinsopp, P.; Hill, R.; Vickerman, J. C. A C<sub>60</sub> Primary Ion Beam System for Time of Flight Secondary Ion Mass Spectrometry: Its Development and Secondary Ion Yield Characteristics. *Anal. Chem.* **2003**, 75, 1754.
66. ION-TOF GmbH, Münster, Germany, [www.iontof.com](http://www.iontof.com).
67. Physical Electronics, Inc., Chanhassen, MN, [www.phi.com](http://www.phi.com).
68. Baldwin, P. M.; Melia, C. D.; Davies, M. C. The Surface Chemistry of Starch Granules Studied by Time-of-Flight Secondary Ion Mass Spectrometry. *J. Cereal Sci.* **1997**, 26, 329.
69. Ameen, A. P.; Short, R. D.; Douglas, C. W. I.; Johns, R.; Ballet, B. A Critical Investigation of Some of the Procedures Employed in the Surgical Use of Titanium. *J. Mater. Sci. Mater. Med.* **1996**, 7, 195.
70. Zhu, Y. J.; Olson, N.; Beebe, T. P. Surface Chemical Characterization of 2.5- $\mu$ m Particulates (PM<sub>2.5</sub>) from Air Pollution in Salt Lake City Using TOF-SIMS, XPS, and FTIR. *Environ. Sci. Technol.* **2001**, 35, 3113.
71. Hattori, A. Measurement of Glass Surface Contamination. *J. Non-Cryst. Solids* **1997**, 218, 196.
72. Spool, A. M. Studies of Adhesion by Secondary-Ion Mass-Spectrometry. *IBM J. Res. Dev.* **1994**, 38, 391.
73. Dietrich, R. Submonolayer Detection of Polymer Additives at the Surface of Industrial Products. *Fresenius J. Anal. Chem.* **1998**, 361, 692.

74. Zimmerman, P. A.; Hercules, D. M.; Rulle, H.; Zehnpfenning, J.; Benninghoven, A. Direct Analysis of Coated and Contaminated Paper Using Time-of-Flight Secondary-Ion Mass-Spectrometry. *Tappi J.* **1995**, 78, 180.
75. Takatsuji, H. Advanced Time-of-Flight Secondary Ion Mass Spectrometry Analyses for Application to TFT-LCD. *Mater. Sci. Semicond. Process.* **2001**, 4, 309.
76. De Witte, H.; De Gendt, S.; Douglas, M.; Conard, T.; Kenis, K.; Mertens, P. W.; Vandervorst, W.; Gijbels, R. Evaluation of Time-of-Flight Secondary Ion Mass Spectrometry for Metal Contamination Monitoring on Si Wafer Surfaces. *J. Electrochem. Soc.* **2000**, 147, 1915.
77. Zanderigo, F.; Ferrari, S.; Queirolo, G.; Pello, C.; Borgini, M. Quantitative TOF-SIMS Analysis of Metal Contamination on Silicon Wafers. *Mater. Sci. Eng. B* **2000**, 73, 173.
78. Chu, P. K.; Schueler, B. W.; Reich, F.; Lindley, P. M. Determination of Trace Metallic Impurities on 200-mm Silicon Wafers by Time-of-Flight Secondary-Ion-Mass Spectroscopy. *J. Vac. Sci. Technol. B* **1997**, 15, 1908.
79. Roche, A.; Marthon, S.; Ple, J. F.; Rochat, N.; Danel, A.; Olivier, M.; Juhel, M.; Tardif, F. Detection and Identification of Organic Contamination on Silicon Substrates. *Solid State Phenom.* **2001**, 76–77, 111.
80. Yamazaki, H.; Tamaoki, M.; Oohashi, M. Quantitative Secondary Ion Mass Spectrometry Analysis of Carbon and Fluorine Impurities on Silicon Wafers Stored in Polymer Carrier Cases. *Jpn. J. Appl. Phys.* **2000**, 39 Part 1, 4744.
81. Guan, J. J.; Gale, G. W.; Bennett, J. Effects of Wet Chemistry Pre-Gate Clean Strategies on the Organic Contamination of Gate Oxides for Metal-Oxide-Semiconductor Field Effect Transistor. *Jpn. J. Appl. Phys.* **2000**, 39 Part 1, 3947.
82. Lazzeri, P.; Lui, A.; Moro, L.; Vanzetti, L. Use of Spin-Coated TXRF Reference Samples for TOF-SIMS Metal Contaminant Quantification on Silicon Wafers. *Surf. Interface Anal.* **2000**, 29, 798.
83. Stevie, F. A.; Roberts, R. F.; McKinley, J. M.; Decker, M. A.; Granger, C. N.; Santiesteban, R.; Hitzman, C. J. Surface Quantification by Ion Implantation Through a Removable Layer. *J. Vac. Sci. Technol. B* **2000**, 18, 483.
84. Stevie, F.A.; Roberts, R.F.; Decker, M.A. Method for Controlled Implantation of Elements into the Surface or Near Surface of a Substrate. U.S. Patent 6,121,624. 1998.
85. Schroder-Oeynhausen, F.; Burkhardt, B.; Fladung, T.; Kotter, F.; Schnieders, A.; Wiedmann, L.; Benninghoven, A. Quantification of Metal Contaminants on GaAs with Time-of-Flight Secondary Ion Mass Spectrometry. *J. Vac. Sci. Technol. B* **1998**, 16, 1002.
86. Claes, M.; De Gendt, S.; Kenens, C.; Conard, T.; Bender, H.; Storm, W.; Bauer, T.; Lagrange, S.; Mertens, P.; Heyns, M. M. A Controlled Deposition of Organic Contamination and the Removal with Ozone Based Cleanings. *Solid State Phenom.* **2001**, 76–77, 223.
87. Martin, A. R.; Baeyens, M.; Hub, W.; Mertens, P. W.; Kolbesen, B. O. Alkaline Cleaning of Silicon Wafers: Additives for the Prevention of Metal Contamination. *Microelectron. Eng.* **1999**, 45, 197.
88. Gillen, G.; Roberson, S. Preliminary Evaluation of an SF<sub>5</sub><sup>+</sup> Polyatomic Primary Ion Beam for Analysis of Organic Thin Films by Secondary Ion Mass Spectrometry. *Rapid Commun. Mass Spectrom.* **1998**, 12, 1303.
89. Szakal, C.; Sun, S.; Wucher, A.; Winograd, N. C<sub>60</sub> Molecular Depth Profiling of a Model Polymer. *Appl. Surf. Sci.* **2004**, 231–232, 183.
90. Cheng, J.; Wucher, A.; Winograd, N. Molecular Depth Profiling with Cluster Ion Beams. *J. Phys. Chem. B* **2006**, 110, 8329.

91. Toyoda, N.; Matsuo, J.; Aoki, T.; Yamada, H.; Fenner, D. B. Secondary Ion Mass Spectrometry with Gas Cluster Ion Beams. *Nucl. Instr. Meth. Phys. Res. B* **2002**, *190*, 860.
92. Rading, D.; Moellers, R.; Cramer, H. G.; Niehuis, E. Dual Beam Depth Profiling of Polymer Materials: Comparison of C<sub>60</sub> and Ar Cluster Ion Beams for Sputtering. *Surf. Interface Anal.* **2013**, *45*, 171.
93. Wehbe, N.; Tabarrant, T.; Brison, J.; Mouhib, T.; Delcorte, A.; Bertrand, P.; Moellers, R.; Niehuis, E.; Houssiau, L. TOF-SIMS Depth Profiling of Multilayer Amino-Acid Films Using Large Argon Cluster Ar<sub>n</sub><sup>+</sup>, C<sub>60</sub><sup>+</sup> and Cs<sup>+</sup> Sputtering Ions: A Comparative Study. *Surf. Interface Anal.* **2013**, *45*, 178.
94. Rabbani, S.; Barber, A. M.; Fletcher, J. S.; Lockyer, N. P.; Vickerman, J. C. TOF-SIMS with Argon Gas Cluster Ion Beams: A Comparison with C<sub>60</sub><sup>+</sup>. *Anal. Chem.* **2011**, *83*, 3793.
95. Moritani, K.; Tanaka, M.; Inui, N.; Mochiji, K. Highly Sensitive Analysis of Surface Contaminants by Ar Gas Cluster SIMS. *Surf. Interface Anal.* **2013**, *45*, 143.
96. Smentkowski, V.; Zorn, G.; Misner, A.; Parthasarathy, G.; Couture, A.; Tallarek, E.; Hagenhoff, B. TOF-SIMS Depth Profiling of Organic Solar Cell Layers Using an Ar Cluster Ion Source. *J. Vac. Sci. Technol. A* **2013**, *31*, 030601.
97. Mouhib, T.; Poleunis, C.; Wehbe, N.; Michels, J. J.; Galagan, Y.; Houssiau, L.; Bertrand, P.; Delcorte, A. Molecular Depth Profiling of Organic Photovoltaic Heterojunction Layers by TOF-SIMS: Comparative Evaluation of Three Sputter Beams. *Analyst* **2013**, *138*, 6801.
98. Fujimoto, H.; Edura, T.; Miyayama, T.; Sanada, N.; Adachi, C. Accurate Measurement of Dopant Concentration in Organic Light-Emitting Diodes by Combining High-performance Liquid Chromatography and TOF-SIMS. *J. Vac. Sci. Technol. B* **2014**, *32*, 030604.
99. Goff, R. F.; Smith, D. P. Surface Composition Analysis by Binary Scattering of Noble Gas Ions. *J. Vac. Sci. Technol.* **1970**, *7*, 72.
100. Boerma, D. O. Surface Physics with Low-Energy Ion Scattering. *Nucl. Instr. Meth. Phys. Res. B* **2001**, *183*, 73.
101. Czanderna, A. W.; Hercules, D. M., Eds., *Ion Spectroscopies for Surface Analysis*; Plenum Press: New York, NY, 1991.
102. Niehus, H.; Heiland, W.; Taglauer, E. Low-Energy Ion Scattering at Surfaces. *Surf. Sci. Rep.* **1993**, *17*, 213.
103. Briggs, D.; Seah, M. P., Eds., *Practical Surface Analysis, Volume 2—Ion and Neutral Spectroscopy*; John Wiley: Chichester, UK, 1992.
104. Rabalais, J. W. Low Energy Ion Scattering and Recoiling. *Surf. Sci.* **1994**, 299–300, 219.
105. Goldberg, E. C.; Monreal, R.; Flores, F.; Brongersma, H. H.; Bauer, P. New Model for Ion Neutralization at Surfaces. *Surf. Sci. Lett.* **1999**, *440*, 875.
106. Taglaner, E. *Ion Spectroscopies for Surface Analysis*; Plenum Press: New York, 1991.
107. Wang, N. P.; Garcia, E. A.; Monreal, R.; Flores, F.; Goldberg, E. C.; Brongersma, H. H.; Bauer, P. Low-Energy Ion Neutralization at Surfaces: Resonant and Auger Processes. *Phys. Rev. A* **2001**, *64*, 12901.
108. Diebold, U.; Pan, J. M.; Madey, T. E. Ultrathin Metal Films on TiO<sub>2</sub> (110): Metal Overlayer Spreading and Surface Reactivity. *Surf. Sci.* **1993**, 287–288, 896.
109. Zhang, L.; Persaud, R.; Madey, T. E. Ultrathin Metal Films on a Metal Oxide Surface: Growth of Au on TiO<sub>2</sub>. *Phys. Rev. B* **1997**, *56*, 10549.
110. Brongersma, H.H.; Maas, A.J.H.; Viitanen, M.M. The Outermost Surface of Polymers Studied by Static Low Energy Ion Scattering. Calipso LEIS Expertise Centre, Eindhoven University of Technology, Eindhoven, The Netherlands. Published at [www.calipso.nl](http://www.calipso.nl). See also [www.tascon.eu](http://www.tascon.eu).

111. Brongersma, H. H.; Carrere-Fontaine, M.; Cortenraad, R.; Denier van der Gon, A. W.; Scanlon, P. J.; Spolveri, I.; Cortigiani, B.; Bardi, U.; Taglauer, E.; Reiter, S.; Labich, S.; Bertrand, P.; Houssiau, L.; Speller, S.; Parascandola, S.; Unlu-Lachnitt, H.; Heiland, W. A Round Robin Experiment of Elemental Sensitivity Factors in Low-Energy Ion Scattering. *Nucl. Instr. Meth. Phys. Res. B* **1999**, *142*, 377.
112. Denier van der Gon, A. W.; Reijme, M. A.; Rumphorst, R. F.; Maas, A. J. H.; Brongersma, H. H. A Combined Time-of-Flight and Electrostatic Analyzer for Low-Energy Ion Scattering. *Rev. Sci. Instrum.* **1999**, *70*, 3910.
113. Epling, W. S.; Mount, C. K.; Hoflund, G. B.; Young, V. Y. Chemical Alteration of Thin Aluminum Films on Aluminum During Hydrogen-Atom Exposures. *Appl. Surf. Sci.* **1998**, *126*, 235.
114. Epling, W. S.; Mount, C. K.; Hoflund, G. B. Study of a Polycrystalline Ni/Cr Alloy V. Hydrogen-Atom Exposure. *Thin Solid Films* **1997**, *304*, 273.
115. Smith, D. P. Analysis of Surface Composition with Low-Energy Backscattered Ions. *Surf. Sci.* **1971**, *25*, 171.
116. Sparrow, G. R.; Foley, J. Quantitative Evaluation and Control of Surface Chemistry Affecting Electronic Devices: ISS and SIMS Surface Analysis. In: *Electrochemical Society Proceedings* **1994**, 94–29, 303.
117. Sparrow, G. R.; Chandler, C. The Role of Surface Chemistry and VCI's in Early Stages of Metal Processing. In: *National Association of Corrosion Engineers Meeting, Corrosion '95, Paper # 489*, Orlando, FL, 1995.
118. Ponjee, M. W. G.; Reijme, M. A.; Denier van der Gon, A. W.; Brongersma, H. H.; Langeveld-Voss, B. M. W. Intermolecular Segregation of Siloxane in P3HT: Surface Quantification and Molecular Surface-Structure. *Polymer* **2002**, *43*, 77.